GAS EXPORT COMPRESSOR VER 1

Operator Training Simulator · export compressor K-100 driven by GT-100

Department of Chemical Engineering, Chulalongkorn University (teaching)

HAZOP REPORT

Hazard and Operability Study of the gas export compressor train K-100

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Teaching document generated from the HAZOP tab of the K-100 simulator. It is not a study of a real plant.

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
A	12-Oct-26	Issued for teaching	K-100 OTS (teaching study)			

HAZOP CLOSE-OUT SUMMARY

Project: Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator · Document KC-V1-SA-HZP-001 rev A · 38 actions, all open at issue. The action party fills "Action taken" and the HAZOP leader changes the status to CLOSED when the close-out sheet (Appendix 3) is signed.

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N1 / **N11** Production header / Blowdown BDV-101 / BDV-112	**R01**	N1.4, N11.4	Relief and flare study: confirm PSV-001 for the blocked-outlet case with the full production (K-100 tripped and PV-102 shut), and the flare header and tip for PV-102 at full production plus simultaneous blowdown (back-pressure on PSV-001 and the BDVs, radiation on the deck).	Process / Safety		OPEN	Risk 10 High
N1 Production header	**R02**	N1.5	Specify the seat-leak class of PV-102 and test it at each shutdown; report flared gas as a loss figure (valued at the training gas price).	Mechanical / Operations		OPEN	Risk 6 Medium
N1 Production header	**R03**	N1.4	Give PSHH-001 its own pressure transmitter: today PT-001 serves the PI-001 indication, PIC-102 and PSHH-001.	Instrument		OPEN	Risk 10 High
N2 Inlet scrubber V-101	**R04**	N2.1	Slug handling: check LV-101 and the closed drain against the largest credible slug and agree a slug warning from production (well tests, pigging, restarts).	Process / Operations		OPEN	Risk 12 High
N2 / **N4** Inlet scrubber V-101 / Intercooler E-101	**R05**	N2.2, N4.4	Gas blow-by protection of the closed drain: low-low level on V-101 and V-102 that closes LV-101 / LV-107, and relief of the drain drum sized for gas blow-by through a wide-open LV.	Process		OPEN	Risk 8 Medium
N2 Inlet scrubber V-101	**R06**	N2.3	Put PSLL-101 on its own transmitter: the trip reads the PIC-101 measurement, so a frozen or drifting PIC-101 transmitter blinds the control loop, the alarm and the trip together.	Instrument		OPEN	Risk 9 Medium
N2 / **N11** Inlet scrubber V-101 / Blowdown BDV-101 / BDV-112	**R07**	N2.7, N11.6	Position switches on HV-101 and HV-102 with an alarm when either is open while the unit is in RUN.	Instrument		OPEN	Risk 9 Medium
N12 Procedures	**R08**	N12.10	Positive isolation for work on the compressor: double block-and-bleed or spades at the ESDVs; ESDV-101 and ESDV-113 leak tests at each proof test.	Operations / Mechanical		OPEN	Risk 8 Medium
N3 / **N5** Section 1 (LP) and UV-103 recycle / Section 2 (HP) and UV-108 recycle	**R09**	N3.2, N5.2	Validate the anti-surge flow measurements: second flow element or a mass-balance deviation alarm, because UIC-103 and UIC-108 each act on one transmitter.	Instrument		OPEN	Risk 12 High
N3 / **N5** / **N12**	**R10**	N3.3, N5.2, N12.3	Alarm when UIC-103 or UIC-108 is not in AUTO with the unit in RUN; manual mode by supervisor access only.	Instrument / Operations		OPEN	Risk 12 High
N3 / **N5** Section 1 (LP) and UV-103 recycle / Section 2 (HP) and UV-108 recycle	**R11**	N3.1, N3.4, N5.1, N5.6	Partial-stroke and stroke-time tests of UV-103 and UV-108; valve-position deviation alarm (ZI against controller output).	Instrument		OPEN	Risk 12 High
N1 / **N2** / **N3** / **N5**	**R12**	N1.2, N2.6, N3.1, N3.2, N5.2	Independent surge detection: surge counter that trips I-1 on repeated surge cycles (for example 3 in 30 s), with the cycles logged for the rotating-equipment engineer.	Instrument / Rotating		OPEN	Risk 12 High
N2 / **N4** / **N6** / **N11**	**R13**	N2.4, N4.5, N6.1, N11.1	Confirm and install PSV-107 (80 barg, V-102) and PSV-112 (138 barg, discharge), on hold on KC-PID-100: size for settle-out with fire, and for a blocked outlet at full GT power.	Process		OPEN	Risk 10 High

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N6 Aftercooler E-102	**R14**	N6.1	Review PSHH-112: set point and delay against PSV-112 (the dynamic check overshoots past the trip setting before the machine unloads); consider a shorter delay or a lower setting.	Instrument / Process		OPEN	Risk 10 High
N6 Aftercooler E-102	**R15**	N6.1, N6.5	ESDV-113: reduce spurious closure (redundant solenoids), proof-test closure time and seat leakage; inspect the NRV at each shutdown.	Instrument / Mechanical		OPEN	Risk 10 High
N7 Export riser and pipeline to the onshore terminal	**R16**	N7.1	Confirm the export pipeline MAOP against the highest discharge pressure (PSV-112 set point); if lower, add pipeline overpressure protection (HIPPS).	Pipeline		OPEN	Risk 12 High
N6 / N7 Aftercooler E-102 / Export riser and pipeline to the onshore terminal	**R17**	N6.2, N7.1, N7.2	Agreement with the onshore terminal on intake restrictions with advance notice; bring PI-900 into the DCS with a high alarm.	Operations / Instrument		OPEN	Risk 12 High
N7 Export riser and pipeline to the onshore terminal	**R18**	N7.3	Riser ESD valve or subsea isolation valve near the platform and pipeline leak detection (pressure/flow balance with the terminal).	Pipeline / Safety		OPEN	Risk 10 High
N1 / N6 / N7	**R19**	N1.7, N6.4, N7.5	Export gas quality: hydrocarbon dew-point limit on TIC-111 and the export gas, and a response plan for an AI-115 high alarm (new wells).	Process / Operations		OPEN	Risk 6 Medium
N1 / N7 Production header / Export riser and pipeline to the onshore terminal	**R20**	N1.8, N7.6	Corrosion management for wet CO_2 (12 mol %): water dew-point monitoring at the inlet, inhibitor, corrosion coupons in the header and the pipeline.	Materials / Process		OPEN	Risk 8 Medium
N4 / N8 Intercooler E-101 / GT-100 driver and fuel gas	**R21**	N4.3, N4.6, N8.5	Fuel-gas conditioning (knock-out, superheater, filter — note 5) with a high-level trip, and a fuel source for start-up and low interstage pressure.	Process		OPEN	Risk 8 Medium
N1 / N8 Production header / GT-100 driver and fuel gas	**R22**	N1.3, N8.2	Operating envelope for the operators: production that K-100 can take against ambient temperature and fuel-gas pressure (GT available power).	Operations / Process		OPEN	Risk 8 Medium
N8 GT-100 driver and fuel gas	**R23**	N8.3	Overspeed protection: confirm the speed probes and voting of SSHH-100 and proof-test it with the GT supplier.	Instrument / Rotating		OPEN	Risk 5 Medium
N3 / N5 / N9	**R24**	N3.7, N5.4, N9.4	Bearing protection: high-high bearing temperature trip or condition monitoring on TI-151 (alarm only today).	Rotating		OPEN	Risk 8 Medium
N9 Lube oil and seal system	**R25**	N9.1, N9.2, N9.3	Lube oil: monitor the readiness of the standby pump (auto-start armed, pump healthy) with a DCS alarm; rundown tank (or DC emergency pump) that lubricates the bearings for the whole coast-down.	Rotating / Electrical		OPEN	Risk 12 High
N9 Lube oil and seal system	**R26**	N9.5, N9.6, N9.7	Extend the HAZOP to the seal-gas panel and the lube-oil console sheets: seal-gas supply at start-up and at settle-out, primary-vent high-flow trip, gas in the oil reservoir.	Process / Rotating		OPEN	Risk 8 Medium
N10 Cooling medium	**R27**	N10.1	Installed spare cooling-medium pump with auto-start on P-160 failure.	Process / Mechanical		OPEN	Risk 12 High
N4 / N6 / N10	**R28**	N4.1, N6.3, N10.3	Second temperature element on the cooler outlets (deviation alarm) so a single transmitter cannot move the cooling medium unseen.	Instrument		OPEN	Risk 9 Medium
N3 / N4 / N5 / N10	**R29**	N3.6, N4.1, N5.3, N5.5, N10.2	Cooler performance monitoring (E-101/E-102 duty and approach, TV positions) and a cleaning programme for the coolers and the seawater exchanger.	Operations		OPEN	Risk 9 Medium

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N10 Cooling medium	**R30**	N10.5, N10.6	Tube-rupture case for E-101 and E-102 (API 521): relief on the cooling-medium side and gas detection at the cooling-medium expansion tank.	Process		OPEN	Risk 8 Medium
N11 Blowdown BDV-101 / BDV-112	**R31**	N11.1	BDV-101/BDV-112 proof test (solenoid, stroke, restriction orifice) and confirmation of the 15-min blowdown with the fire case.	Instrument / Safety		OPEN	Risk 10 High
N2 / **N11** Inlet scrubber V-101 / Blowdown BDV-101 / BDV-112	**R32**	N2.8, N11.3	Low-temperature materials: confirm flare lines, BDV/HV-102 outlets and HV-101 for the expansion cooling (MDMT −46 °C, note 3, TBC).	Materials		OPEN	Risk 8 Medium
N11 Blowdown BDV-101 / BDV-112	**R33**	N11.7, N11.8	Flare knock-out drum, flare pilot monitoring and liquid carry-over protection (outside this node set; confirm in the flare study).	Process		OPEN	Risk 8 Medium
N11 / **N12** Blowdown BDV-101 / BDV-112 / Procedures	**R34**	N11.5, N12.9	ESD-2 criteria in the emergency procedure (when to blow down) and automatic ESD-2 from confirmed fire detection.	Operations / Safety		OPEN	Risk 8 Medium
N7 / **N8** / **N12**	**R35**	N7.2, N8.1, N12.7, N12.8	Automatic production cut-back signal to the wells on a K-100 trip, and ramp-rate limits for production changes agreed with production control.	Operations / Process		OPEN	Risk 12 High
N2 / **N12** Inlet scrubber V-101 / Procedures	**R36**	N2.5, N12.3, N12.5	Restart procedure: vent the settled-out loop with HV-102 before HV-101 or ESDV-101 is opened; simulator training on restart and loading.	Operations		OPEN	Risk 9 Medium
N1 / **N6** / **N8** / **N9**	**R37**	N1.9, N6.7, N8.7, N9.6	Cover gas release, fire and gas detection and the ESD response of the compressor module in the fire-and-gas study (not on KC-PID-100).	Safety		OPEN	Risk 10 High
N1 / **N3** Production header / Section 1 (LP) and UV-103 recycle	**R38**	N1.1, N1.2, N3.1	Feed-forward of the production flow FI-001 to UIC-103/UIC-108 (or rate-of-change limits) so a fast production cut does not drive the margins below the fast-opening line.	Instrument / Process		OPEN	Risk 12 High

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Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator

Department of Chemical Engineering, Chulalongkorn University (teaching)

	Role	Name	Signature	Date
Prepared by	HAZOP scribe			
Reviewed by	HAZOP leader (chair)			
Approved by	Asset / project manager			

This report records a teaching HAZOP of the export compression train drawn in the simulator, prepared by the K-100 OTS (teaching study). Names, signatures and dates are filled in by the class team that runs the study; the risk matrix is an example for teaching, not a corporate matrix.

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Executive summary

A HAZOP of the gas export compressor train — the production header with PV-102 and PSV-001, the inlet scrubber V-101, the two process sections of K-100 with their anti-surge recycles UV-103 and UV-108, the intercooler E-101 and interstage scrubber V-102 with the fuel-gas offtake, the aftercooler E-102 and ESDV-113, the export riser and pipeline to the onshore terminal, the GT-100 driver, lube oil and seals, the cooling medium, blowdown and flare, and the pressurising, start-up, loading, settle-out, restart and blowdown procedures — was carried out with the IEC 61882 guide-word method. The unit was divided into **12 nodes**; **87 deviations** were recorded and **38 recommendations** raised; **10 deviations** were also run on the simulator model (Section 5.3).

Nodes	Deviations	Recommendations	Very high	High	Medium	Low
12	87	38	0	14	66	7

Risk is ranked with the existing safeguards credited (Section 3.4). No deviation was ranked very high; the 14 high-ranked deviations (N1.4, N2.1, N3.1, N3.2, N3.3, N5.2, N6.1, N6.7, N7.1, N7.3, N8.1, N9.2, N10.1, N11.1) are those where an instrumented trip, a relief valve on hold or the operator is the main protection — header and discharge overpressure, liquid carry-over into section 1, surge with the anti-surge loop degraded, line-pack and rupture of the export pipeline, loss of the driver, lube oil or cooling medium, and a blowdown that fails. They are carried to the LOPA, the SRS of the SIFs (KC-V1-IN-SRS-001) and the cause and effect diagram (KC-V1-IN-CE-001).

Key findings (dynamic check on the simulator)

- **Driver lost (N8):** the trip is immediate and both recycles open: no surge cycle in the coast-down (at rest at 07:10). PV-102 holds the header at 32.00 barg but flares up to 243 t/h — 109 t of gas in 30 min. The process is safe; the loss is the flare (R35).
- **Production falls 55 % (N1):** no trip; the section 1 margin dips to 4.8 % — below the 10 % control line, into the fast-opening band (5 %) — before the speed comes down to 80.0 %. The anti-surge loops are the only thing between a fast production cut and surge (R38).
- **Cooling medium lost (N10):** P-160 STOP at once, TIC-106 HI at 00:28, TI-109 HI at 00:47 and TSHH-109 trips K-100 at 01:28. About a minute and a half from pump stop to trip: the operator cannot restore cooling in time, so a spare with auto-start is the practical layer (R27).
- **UV-108 fails open (N5):** no trip at all. Export falls to 0 MMSCFD at first, GT-100 hits its power limit in seconds, the speed reaches 105.5 % (below SSHH-100 at 110 %) and PV-102 flares up to 189 t/h — 44 t in 30 min. An operability and loss case the alarms announce clearly (R11).
- **Export blocked by ESDV-113 (N6):** PI-112 HI at 00:04, PSHH-112 trips at 00:10, yet the discharge still overshoots to 135.5 barg — only a few bar below the 138-barg PSV-112 that is on hold on the P&ID. Without the trip the model passes 138 barg in about 11 s (R13, R14).
- **Lube oil lost with a failed standby (N9):** PI-150 LO at 00:02 and PSLL-150 trips at 00:05 — the trip is fast, but the train turns until 07:15 with no oil pressure. The SIF cannot protect the bearings during the coast-down; a rundown tank can (R25).
- **Arrival pressure +30 bar (N7):** PI-112 HI at 02:14 and PSHH-112 at 06:40: about 4 min for the operator to call the terminal or cut production. The terminal's pressure is visible only as telemetry, with no alarm (R17).
- **Liquid slug (N2):** LIC-101 HI at 00:34 and LSHH-101 trips at 01:05 with the level at 87 %: about 31 s between alarm and trip — no time for an operator. LSHH-101 is the real protection of section 1 (R04).
- **Anti-surge in MAN during a production cut (N3):** section 1 surges from 00:22 (9 cycles, 1 in section 2); vibration HI follows, and TSHH-104 finally trips at 00:54 with TI-104 at 156.0 °C. The trip ends the surge only indirectly, through the heat it creates (R10, R12).
- **Flare valve shut when K-100 trips (N1):** the header climbs to 36.00 barg; PSHH-001 shuts the wells at 00:11 and PSV-001 lifts at 00:11. Both last layers act, but PSHH-001 reads the same transmitter as PIC-102, so one failed transmitter defeats the flare control and the shut-in together (R01, R03).

Actions by action party

Action party	Number	Action No.
Process	15	R01, R04, R05, R13, R14, R19, R20, R21, R22, R26, R27, R30, R33, R35, R38
Instrument	14	R03, R06, R07, R09, R10, R11, R12, R14, R15, R17, R23, R28, R31, R38
Operations	11	R02, R04, R08, R10, R17, R19, R22, R29, R34, R35, R36
Rotating	5	R12, R23, R24, R25, R26
Safety	5	R01, R18, R31, R34, R37
Mechanical	4	R02, R08, R15, R27
Materials	2	R20, R32
Pipeline	2	R16, R18
Electrical	1	R25

An action shared by two parties (e.g. Instrument / Operations) is counted under both.

1 Introduction

1.1 Purpose

The HAZOP identifies how the export compression train can deviate from its design intent, what the deviations lead to, whether the existing safeguards are adequate, and what should be changed. It also finds operability problems: deviations that harm no one but leave the platform unable to export what the wells produce, so that the gas is flared, or that damage the compressor, its seals or bearings. The study feeds the safety layers, LOPA and SRS of the SIFs (KC-V1-IN-SRS-001), and the cause and effect diagram KC-V1-IN-CE-001.

1.2 The unit

Production gas from the HP separator (200.0 MMSCFD = 61.75 kg/s, MW 22.27) enters the production header at 30.30 barg through the well shut-in valve XV-001; PIC-102 keeps the header at or below its set point by flaring through PV-102 (fail open), and PSV-001 (36 barg) relieves to the flare. Through ESDV-101 the gas reaches the inlet scrubber V-101 (30.00 barg, PIC-101). The centrifugal compressor K-100 has two process sections on one shaft: section 1 raises the pressure to 60.9 barg (ratio 2.00, discharge 105.2 °C, polytropic efficiency 0.685); the intercooler E-101 (7.30 MW) cools it to 40 °C and the interstage scrubber V-102 removes 0.11 kg/s of condensate and supplies the fuel gas; section 2 raises it to 121.6 barg (ratio 2.00, 115.7 °C, efficiency 0.655), and the aftercooler E-102 (7.73 MW) cools it to 45 °C into the discharge header at 120.9 barg. The export of 195.2 MMSCFD (60.25 kg/s) passes the non-return valve and ESDV-113 into the subsea pipeline (120.1 barg at the riser, 70 barg on arrival onshore). Each section has its own anti-surge loop (UIC-103 / UIC-108 → UV-103 / UV-108, fail open, cooled recycle; margins 61.1 % and 61.4 % at design). The two-shaft gas turbine GT-100 gives 16.69 MW to the shaft of 19.86 MW available at 30 °C ambient, at 99.9 % speed, burning 4.97 t/h of fuel gas; PIC-101 sets the speed through SIC-100 in cascade. The cooling medium (P-160, 30 °C supply) serves both coolers through TV-106 and TV-111 (fail open); the lube-oil system has the main pump P-150A and the standby P-150B with auto-start.

Protection: the I-1 compressor trip (XV-100 fuel, ESDV-101 and ESDV-113 shut, both recycles to 100 %) acts on LSHH-101 V-101 liquid level high-high; LSHH-107 V-102 liquid level high-high; TSHH-104 section 1 discharge ≥ 150 °C; TSHH-109 section 2 discharge ≥ 150 °C; VSHH-105 section 1 vibration ≥ 90 µm; VSHH-110 section 2 vibration ≥ 90 µm; PSLL-150 lube oil pressure ≤ 1.2 barg; SSHH-100 overspeed ≥ 110 %; PSHH-112 discharge ≥ 132 barg; PSLL-101 suction ≤ 20 barg; GT-100 trip / flame-out; ESD-2 adds the blowdown valves BDV-101 / BDV-112 to the flare and shuts the wells (XV-001); PSHH-001 (34.5 barg) shuts the wells on header overpressure. After a trip the loop settles out at about 70.4 barg (Process-tab estimate; 69.4 barg in the dynamic model).

1.3 Objectives

- Identify hazards to people, the environment and the installation from deviations of each node and each procedure step.
- Identify operability problems: export short of production (gas flared), surge and stonewall, a tripped train that cannot be restarted, damaged seals and bearings.
- Judge the existing safeguards and rank the risk with the example matrix.
- Raise recommendations and pass the high-ranked scenarios to LOPA.
- Check selected deviations dynamically on the simulator model (Section 5.3).

2 Scope

Boundaries: scope from XV-001 on the production header (gas from the HP separator) through ESDV-101, V-101, K-100 sections 1 and 2 with UV-103/UV-108, E-101/V-102 with the fuel-gas offtake, E-102, the NRV and ESDV-113 to the export riser and the onshore arrival interface (PI-900); GT-100 with its fuel, the lube-oil and seal systems, the cooling-medium loop, PV-102, PSV-001, BDV-101/112 and HV-102 to the flare. Excluded: wells and HP separator, onshore terminal, seawater system, flare knock-out drum and tip internals, the seal-gas panel and lube-oil console sheets (separate drawings), fire and gas detection. Objectives, team, risk matrix.

A node is a part of the plant with one clear design intent. Boundaries are put where the intent changes: the header collects and limits pressure, V-101 removes liquid, each section raises the pressure about twofold while its recycle keeps it out of surge, E-101 and V-102 cool and dry the interstage gas (and supply the fuel), E-102 and ESDV-113 deliver to the pipeline, which carries the gas ashore. The driver, lube oil and seals, cooling medium and the blowdown/flare system each have their own intent. Procedures get their own node N12 because pressurising, starting, loading, venting the settled-out loop and blowing down create deviations that steady running does not.

2.1 Nodes

Node	Description	Design intent	Design conditions	Equipment / instruments
N1	Production header, PV-102 / PSV-001 to flare	Receive dehydrated production gas (61.75 kg/s = 200 MMSCFD at 35 °C) from the HP separator through XV-001 at about 30.3 barg and pass it to K-100 through ESDV-101; keep the header at or below 32 barg by flaring the excess through PV-102 whenever K-100 cannot take it.	Production 200.0 MMSCFD (61.75 kg/s) at 30.30 barg, 35 °C; PV-102 closed, flare 0.0 t/h; wells shut-in pressure 59 barg (model); MW 22.27	Production header DN500-PG-101 (150 m³ lumped), XV-001 well shut-in valve, PV-102 to flare, PSV-001, flare header FL-100 *FI-001, PI-001 / PT-001, PIC-102, PSHH-001, FI-109*
N2	Inlet scrubber V-101, ESDV-101 and HV-101	Isolate the compressor loop at ESDV-101 (pressurise through HV-101 for start-up), remove liquids from the suction gas in V-101 (LIC-101 40 %, LV-101 fail closed to the closed drain) and deliver dry gas to section 1 at 30 barg (PIC-101).	Suction 30.00 barg, 34.4 °C; level 40 %, liquids 0.22 kg/s to the closed drain; settle-out after a trip ≈ 70.4 barg (Process-tab estimate)	ESDV-101 with HV-101 bypass, V-101 inlet scrubber (25 m³ gas, mesh), LV-101 to closed drain, BDV-101 and HV-102 connections *PIC-101, PSLL-101, TI-101, LIC-101, LSHH-101, HIC-101*
N3	Section 1 (LP) and UV-103 recycle	Compress 61.75 kg/s from 30 to about 61 barg (r ≈ 2.0, Ep 0.685) at about 100 % speed, and keep the operating point at least 10 % from the surge line by recycling cooled gas from the V-102 outlet through UV-103 (fail open).	Q 2.116 m³/s at suction, q 0.999, r 1.997, Ep 0.685, discharge 60.9 barg / 105.2 °C, margin 61 %, UV-103 0 %, throat Mach 0.74	K-100 section 1 (4 impellers), FT-103, UV-103 (DN300-PG-104 LP recycle) *FI-103, UIC-103, PI-104, TI-104, TSHH-104, VI-105, VSHH-105*
N4	Intercooler E-101, interstage scrubber V-102 and fuel-gas offtake	Cool the section 1 discharge (about 106 °C) to 40 °C in E-101 (TIC-106 → TV-106, fail open), separate the condensate in V-102 (LIC-107 40 %, LV-107 fail closed), and supply section 2 at about 60.4 barg and GT-100 with fuel gas (1.38 kg/s).	E-101 duty 7.30 MW, TV-106 55 %, gas out 40.0 °C; V-102 60.41 barg, level 40 %, condensate 0.11 kg/s; fuel 1.38 kg/s	E-101 intercooler (gas / cooling medium), V-102 interstage scrubber (40 m³ gas node with E-101), LV-107 to closed drain, fuel-gas offtake to the GT skid *TIC-106, PI-107, LIC-107, LSHH-107*
N5	Section 2 (HP) and UV-108 recycle	Compress 60.26 kg/s from about 60.4 to 121.6 barg (r ≈ 2.0, Ep 0.655) and keep at least 10 % surge margin by recycling cooled gas from after E-102 to before E-101 through UV-108 (fail open).	Q 0.998 m³/s at suction, q 1.001, r 1.997, Ep 0.655, discharge 121.6 barg / 115.7 °C, margin 61 %, UV-108 0 %	K-100 section 2 (4 impellers), UV-108 (DN250-PG-107 HP recycle) *FI-108, UIC-108, PI-109, TI-109, TSHH-109, VI-110, VSHH-110, TI-151*
N6	Aftercooler E-102, discharge header and ESDV-113	Cool the section 2 discharge (about 116 °C) to 45 °C in E-102 (TIC-111 → TV-111, fail open) and deliver export gas at about 121 barg through the NRV, ESDV-113 and FE-114 to the riser; PSHH-112 protects against a blocked outlet.	E-102 duty 7.73 MW, TV-111 49 %, export 45.0 °C at 120.9 barg; FI-114 195.2 MMSCFD	E-102 aftercooler, discharge header DN400-EG-108 (30 m³), NRV, ESDV-113, FE-114, BDV-112 connection *TIC-111, PI-112, PSHH-112, FI-114, PI-114, AI-115*
N7	Export riser and pipeline to the onshore terminal	Transport about 60.26 kg/s (195 MMSCFD) at about 120 barg and 45 °C through the export riser and 36-in pipeline (lumped teaching volume 450 m³) to the onshore terminal at about 70 barg (PI-900), inside the gas specification (AI-115).	Inlet 120.1 barg, arrival 70.0 barg, 195.2 MMSCFD, gas velocity 0.9 m/s, MW 22.27	Export riser, subsea pipeline (Weymouth-type line), onshore receiving terminal (outside the scope) *FI-114, PI-114, PI-900 (telemetry), AI-115*

Node	Description	Design intent	Design conditions	Equipment / instruments
N8	GT-100 driver and fuel gas	Drive K-100 directly at 9500 rpm = 100 % (SIC-100 range 70–105 %) with up to 19.86 MW available at 30 °C (22 MW ISO, −0.65 %/°C above 15 °C), burning fuel gas from V-102 through XV-100 and FV-100 (fail closed).	Speed 99.9 %, shaft 16.69 MW of 19.86 MW available (margin 3.16 MW), FV-100 84 %, fuel 4.97 t/h, exhaust 574 °C, ambient 30 °C	GT-100 two-shaft industrial gas turbine, fuel-gas skid, XV-100, FV-100, exhaust *SIC-100, SSHH-100, JI-100, FI-100, TI-100, TI-099*
N9	Lube oil and seal system	Supply lube oil at about 3.0 barg to the K-100 and GT bearings (P-150A main, P-150B standby auto-starting at 1.8 barg with its selector in AUTO) while running and through the coast-down; keep the process gas inside the casing with the shaft seals (seal-gas panel on a separate sheet).	PI-150 3.00 barg, bearing 74.5 °C; trip at 1.2 barg, standby start at 1.8 barg	Lube-oil console (P-150A/B, reservoir, coolers, filters), shaft seals and seal-gas panel (separate sheet) *PI-150, PSLL-150, TI-151*
N10	Cooling medium: P-160, TV-106, TV-111	Circulate cooling medium at 30 °C (TI-160) with P-160 through E-101 and E-102 (TV-106 and TV-111 fail open, about half open at design) and reject 15.2 MW to seawater.	Supply 30.0 °C; TV-106 55 % (≈ 43 kg/s), TV-111 49 % (≈ 40 kg/s); duties 7.30 + 7.73 MW	P-160, cooling-medium supply and return headers, TV-106, TV-111, seawater exchanger (separate sheet) *TI-160, TIC-106, TIC-111*
N11	Blowdown BDV-101 / BDV-112, HV-102 and flare	On ESD-2 depressurise the compressor loop through BDV-101 (suction side) and BDV-112 (discharge), RO 45 mm and choked, to 6.9 barg within 15 min (API 521); after a shutdown vent the settled-out loop with HV-102; carry PV-102, PSV-001, BDV and HV-102 flows safely to the flare.	Loop inventory to blow down: S1 25 + D1 40 + D2 30 m³; settle-out ≈ 70.4 barg; model blowdown to 6.9 barg in 6.1 min, flare line down to −21 °C	BDV-101, BDV-112 (fail open, RO 45 mm), HV-102, flare header DN500-FL-100, flare tip *FI-109, HIC-102*
N12	Procedures: pressurising, GT start, loading, settle-out, restart, blowdown	Bring K-100 from a depressurised loop to full load and back safely: pressurise with HV-101, open ESDV-101, start GT-100 on full recycle, load with SIC-100 in cascade, stop with ESD-1 after the wells are cut back, vent the settled-out loop with HV-102 before a restart, and blow down with ESD-2 in an emergency.	Start-up limit: GT at minimum governor speed 70 %; reset needs the train below 5 %; ESDV-101 ΔP < 3 bar; blowdown to 6.9 barg in 6.1 min (model)	Whole unit; Exercises tab procedures (cold start-up, normal shutdown, ESD with blowdown, GT trip restart) *SIS screen (reset, ESD-1, ESD-2, ESDV commands, GT start), HIC-101, HIC-102*

2.2 Node map

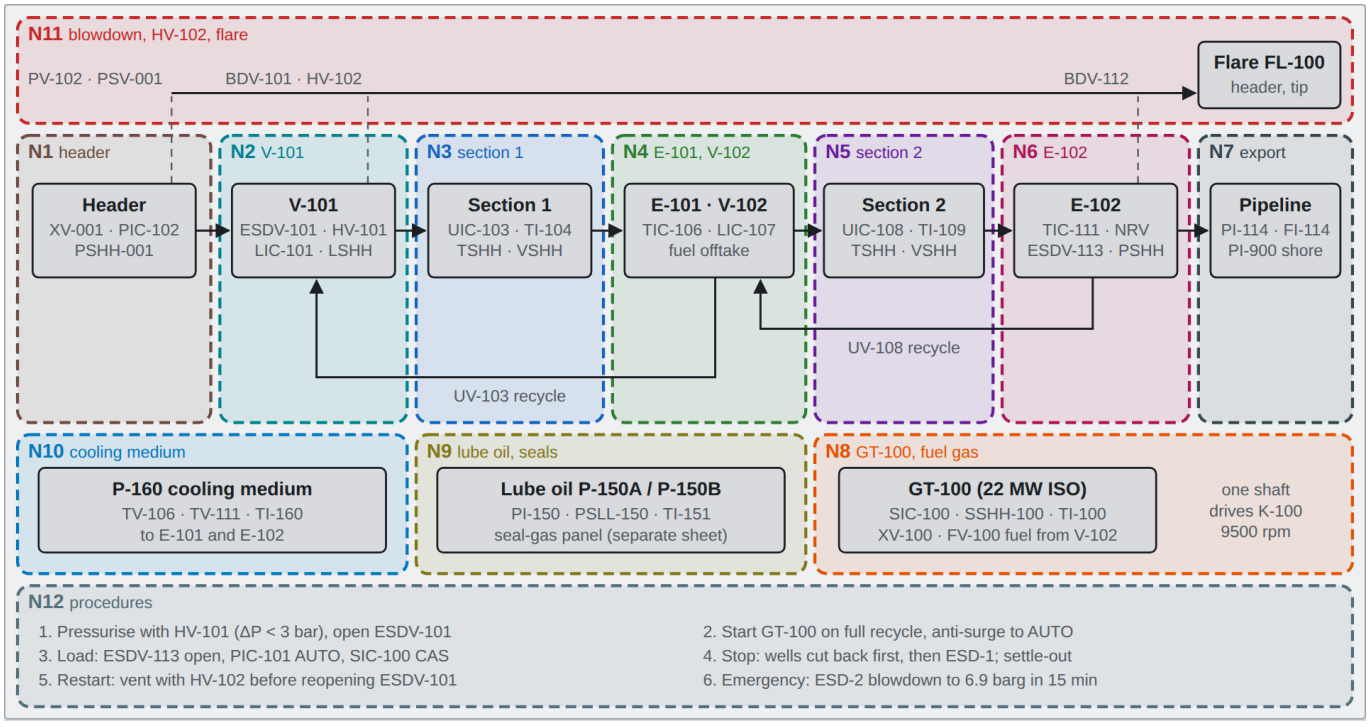


Figure 1. The export compression train divided into HAZOP nodes (N12, the procedures node, covers the whole train; its steps are listed in the bottom band).

3 Methodology

HAZOP is a structured, team-based examination of a defined system. Each part of the design is compared with its **design intent**; **guide words** combined with process **parameters** generate **deviations**; for each credible deviation the team records causes, consequences, existing safeguards and, where the risk is not acceptable, recommendations. It finds hazards (harm to people, environment, assets) and operability problems (the platform cannot export what the wells produce and flares it instead). On a centrifugal export compressor the hazards are dominated by stored energy: high-pressure gas inventories that settle out after every trip, surge when the flow falls below the peak of the head curve, a discharge that packs within seconds when the outlet is blocked, and a rotor that keeps turning for minutes after the driver stops.

3.1 References

Reference	What it contributes
IEC 61882:2016 *Hazard and operability studies (HAZOP studies) – Application guide*	Definitions, the four phases (definition, preparation, examination, documentation and follow-up), guide words, team roles, recording
CCPS *Guidelines for Hazard Evaluation Procedures* (3rd ed.)	Worksheet format, node selection, procedural HAZOP, "consequence without safeguards", risk ranking, links to LOPA
IEC 61511 (functional safety, SIS)	How HAZOP scenarios feed LOPA and SIL targets for the I-1 trips (PSHH-112, LSHH-101/107, TSHH-104/109, PSLL-150, SSHH-100), PSHH-001 and proposed ones (surge counter, independent PSLL-101)
API 617 (centrifugal compressors), API 614 (lube-oil systems), API 692 (dry gas seals), API 616 (gas turbines)	Surge and anti-surge, settle-out, lube-oil standby and coast-down, seal-gas supply, overspeed protection used as the basis for safeguards
API 521 / ISO 23251 (pressure relief and depressuring), API RP 14C / ISO 10418 (offshore safety analysis)	Relief cases (blocked outlet, fire, tube rupture), blowdown to 6.9 barg within 15 min, low-temperature flare lines, safety analysis tables for production facilities
ISO 31000 / owner's risk matrix	Risk criteria used to decide whether safeguards are adequate
Thailand: Department of Industrial Works regulation on hazard identification, risk assessment and risk-management plans (B.E. 2543); petroleum safety rules of the Department of Mineral Fuels for offshore production (check the current notification)	Legal requirement to identify hazards (HAZOP is an accepted method) and to keep risk-management plans
OSHA 29 CFR 1910.119 (PSM), where applicable	Process hazard analysis for covered processes, revalidated at least every five years

3.2 The four phases

• **Definition**: scope from XV-001 on the production header (gas from the HP separator) through ESDV-101, V-101, K-100 sections 1 and 2 with UV-103/UV-108, E-101/V-102 with the fuel-gas offtake, E-102, the NRV and ESDV-113 to the export riser and the onshore arrival interface (PI-900); GT-100 with its fuel, the lube-oil and seal systems, the cooling-medium loop, PV-102, PSV-001, BDV-101/112 and HV-102 to the flare. Excluded: wells and HP separator, onshore terminal, seawater system, flare knock-out drum and tip internals, the seal-gas panel and lube-oil console sheets (separate drawings), fire and gas detection. Objectives, team, risk matrix.

• **Preparation**: collect KC-PID-100, the heat and mass balance (Process tab), the compressor maps and the outlet-temperature method (Theory tab), the trip, permissive and alarm lists, the start-up, shutdown, restart and blowdown procedures, datasheets; divide the plant into nodes; write each node's design intent; choose parameters and guide words.

• **Examination**: for each node → state the design intent → choose a parameter → apply each guide word → is the deviation meaningful? → causes → consequences assuming the safeguards fail → existing safeguards → rank the risk → recommend → next guide word. Record every meaningful deviation, including those with adequate safeguards ("full recording"), so the reasoning can be audited.

• **Documentation and follow-up**: worksheets, recommendation register with owners and dates, action close-out, and revalidation after changes (MOC) or periodically.

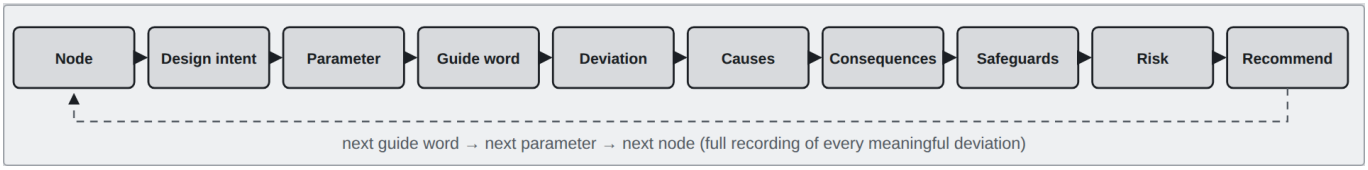


Figure 2. HAZOP examination loop.

3.3 Guide words and parameters

Guide word	Meaning (IEC 61882)	Rows	Example from this study
NO / NOT	Complete negation of the intent	11	Flow NO: No production gas (N1.1)
MORE	Quantitative increase	27	Flow MORE: Production above the design rate (N1.3)
LESS	Quantitative decrease	19	Flow LESS: Production falls fast (N1.2)
AS WELL AS	Qualitative increase: something extra present	8	Composition AS WELL AS: Free water with CO_2-rich gas (N1.8)
PART OF	Qualitative decrease: only part of the intent achieved	2	Sequence PART OF: Ignition without a complete purge (N8.6)
REVERSE	Logical opposite of the intent	2	Flow REVERSE: Settled-out gas back into the header (N2.5)
OTHER THAN	Complete substitution: something else happens	9	Composition OTHER THAN: Richer (heavier) gas (N1.7)
EARLY	Relative to clock time	1	Timing EARLY: Loaded before the anti-surge loops are in AUTO (N12.3)
LATE	Relative to clock time	4	Timing LATE: Recycle opens too slowly on a trip (N5.6)
BEFORE	Relative to order or sequence	2	Sequence BEFORE: ESDV-101 opened before pressurising (N12.1)
AFTER	Relative to order or sequence	2	Sequence AFTER: GT started before the recycles are open (N12.2)

Parameters used: flow, pressure, temperature, level, composition, vibration, speed, power (driver), measurement, control, integrity (containment), ignition (flare), and for procedures: sequence, timing, restart, shutdown, blowdown, maintenance. A deviation is recorded only if it is physically meaningful ("REVERSE temperature" is not; REVERSE flow from the pipeline or from the settled-out loop is).

3.4 Risk ranking

S \ L	L1	L2	L3	L4	L5
S5	5 (1)	10 (5)	15	20	25
S4	4 (1)	8 (20)	12 (8)	16	20
S3	3 (1)	6 (16)	9 (9)	12 (1)	15
S2	2	4 (3)	6 (17)	8 (3)	10
S1	1	2	3 (2)	4	5

Numbers in brackets: how many deviations of this study fall in the cell. Risk = S × L (the same matrix ranks the HAZID register).

1–4 Low	acceptable
5–9 Medium	reduce if practicable (ALARP)
10–16 High	recommendation required, consider LOPA
20–25 Very high	not acceptable

S	Severity (worst credible consequence, no credit for safeguards)
1	No injury; short upset, no flaring
2	First aid; production flared or export reduced for hours; minor equipment damage
3	Lost-time injury; compressor down for days (export lost); reportable flaring; small gas release
4	Permanent disability or single fatality; gas release with fire or explosion on the module; major machine damage (rotor, seals, bearings)
5	Multiple fatalities; escalation (riser or pipeline rupture, high-pressure jet fire, platform-wide fire)

L	Likelihood with the existing safeguards
1	Remote: not expected in the life of the plant ($< 10^{-4}$ /y)
2	Unlikely: known in industry (10^{-4}–10^{-3} /y)
3	Possible: could happen in the plant life (10^{-3}–10^{-2} /y)
4	Likely: several times in the plant life (10^{-2}–10^{-1} /y)
5	Frequent: yearly or more

Consequences are written assuming the safeguards fail (worst credible case). S ranks that consequence; L ranks its likelihood with the listed safeguards working as designed. Safeguard settings are read from the simulator's own trip, delay, set-point and alarm tables; numbers quoted in a consequence come from runs on copies of the model (trips disabled where the text says "no trip"). "Fault library" marks a cause that the Instructor station can inject. PSV-107 and PSV-112 are listed but not credited: KC-PID-100 note 3 holds them.

4 HAZOP sessions

4.1 Schedule

The study is run in class as a teaching HAZOP. A typical plan is four sessions; the team records the dates and attendance on this page.

Session	Nodes	Date	Remark
1	N1 – N3: Production header; Inlet scrubber V-101; Section 1 (LP) and UV-103 recycle		
2	N4 – N6: Intercooler E-101; Section 2 (HP) and UV-108 recycle; Aftercooler E-102		
3	N7 – N9: Export riser and pipeline to the onshore terminal; GT-100 driver and fuel gas; Lube oil and seal system		
4	N10 – N12: Cooling medium; Blowdown BDV-101 / BDV-112; Procedures; review of the recommendations		

4.2 Team

Role	Responsibility	Name	Attended sessions
HAZOP leader (chair)	Independent of the design; runs the method, keeps the team on the node, decides when a deviation is covered		
Scribe (recorder)	Writes the worksheet live, reads back recommendations		
Process engineer (designer)	Design intent, heat and mass balance, settle-out, relief and blowdown cases		
Operations representative (platform)	How the train is really run: loading, turndown during well work, restart after a trip, flaring practice, communication with production control and the onshore terminal		
Instrument / control engineer	Anti-surge and capacity control, fail positions, trips and permissives, which transmitter each trip reads		
Rotating-equipment engineer and GT vendor	Surge, seals, bearings, lube oil and coast-down, overspeed, GT fuel quality and start sequence		
Safety / risk engineer	Risk matrix, fire and gas, LOPA follow-up, regulations		
Specialists as needed	Pipeline engineer (MAOP, riser, line-pack), materials (CO_2 corrosion, low temperature), onshore terminal representative		

4.3 Documents reviewed

Document	Title / content
KC-PFD-100 rev A	Process flow diagram of the export compression train with the stream numbers of the Process tab (PFD tab; Appendix 2)
KC-PID-100 rev A	P&ID of the production header, V-101, K-100 sections 1 and 2, E-101, V-102, E-102, ESDV-113, GT-100, lube oil, cooling medium, blowdown and flare, with notes (P&ID tab; Appendix 2)
KC-V1-IN-CE-001 rev R0	Cause and effect diagram of the I-1 / ESD-2 trips, start permissives and alarms (C&E matrix tab)
KC-V1-IN-SRS-001 rev R0	Safety requirements specification of the SIFs (SRS of SIF tab)
The OTS model (Gas Export Compressor Ver 1)	Process tab (stream table, heat, mass and power balances, block flow diagram), Theory tab (outlet temperature, sonic flow), Compressor curves, Design of the unit operation, Control loop design, Safety system (SIS) tab (trips, delays, start permissives), Workbook (alarm list), the start-up, shutdown, restart and blowdown procedures (Exercises); the dynamic model used for Section 5.3

4.4 Nodes reviewed and guide-word coverage

Node	NO / NOT	MORE	LESS	AS WELL AS	PART OF	REVERSE	OTHER THAN	EARLY	LATE	BEFORE	AFTER	Rows	Parameters
N1	1	3	2	1	·	·	2	·	·	·	·	9	Flow, Pressure, Temperature, Composition, Integrity
N2	1	2	3	1	·	1	·	·	·	·	·	8	Level, Pressure, Flow, Temperature
N3	1	3	1	·	·	·	2	·	·	·	·	7	Flow, Measurement, Control, Temperature, Vibration
N4	·	3	3	·	·	·	·	·	·	·	·	6	Temperature, Level, Pressure
N5	·	4	1	·	·	·	·	·	1	·	·	6	Flow, Temperature, Vibration, Pressure, Timing
N6	1	2	2	·	·	1	1	·	·	·	·	7	Flow, Pressure, Temperature, Integrity
N7	1	1	2	1	·	·	1	·	·	·	·	6	Pressure, Flow, Integrity, Composition
N8	1	2	1	1	1	·	1	·	·	·	·	7	Power, Speed, Temperature, Composition, Sequence, Integrity
N9	2	2	2	1	·	·	·	·	·	·	·	7	Pressure, Power, Temperature, Composition, Flow
N10	1	3	1	1	·	·	·	·	·	·	·	6	Flow, Temperature, Pressure, Composition
N11	2	2	1	2	·	·	·	·	1	·	·	8	Flow, Temperature, Pressure, Timing, Composition, Ignition
N12	·	·	·	·	1	·	2	1	2	2	2	10	Sequence, Timing, Restart, Shutdown, Blowdown, Maintenance

A dot means that guide word gave no meaningful deviation for any parameter in the node (it was still considered). EARLY / LATE / BEFORE / AFTER apply mainly to the procedure node N12 and to the timing of protective actions.

5 Findings and recommendations

5.1 Risk profile

Node	Deviations	High	Medium	Low	Recommendations
N1 Production header	9	1	7	1	R01, R02, R03, R12, R19, R20, R22, R37, R38
N2 Inlet scrubber V-101	8	1	7	0	R04, R05, R06, R07, R12, R13, R32, R36
N3 Section 1 (LP) and UV-103 recycle	7	3	4	0	R09, R10, R11, R12, R24, R29, R38
N4 Intercooler E-101	6	0	6	0	R05, R13, R21, R28, R29
N5 Section 2 (HP) and UV-108 recycle	6	1	5	0	R09, R10, R11, R12, R24, R29
N6 Aftercooler E-102	7	2	3	2	R13, R14, R15, R17, R19, R28, R37
N7 Export riser and pipeline to the onshore terminal	6	2	3	1	R16, R17, R18, R19, R20, R35
N8 GT-100 driver and fuel gas	7	1	5	1	R21, R22, R23, R35, R37
N9 Lube oil and seal system	7	1	6	0	R24, R25, R26, R37
N10 Cooling medium	6	1	5	0	R27, R28, R29, R30
N11 Blowdown BDV-101 / BDV-112	8	1	6	1	R01, R07, R13, R31, R32, R33, R34
N12 Procedures	10	0	9	1	R08, R10, R34, R35, R36
Total	87	14	66	7	38

High-ranked deviations

Ref	Deviation	Consequence (no credit for safeguards)	S×L	Rec.
N1.4	Pressure MORE: Header overpressure when K-100 stops	Full production into a closed 150 m³ header: the pressure climbs toward the wells' shut-in pressure (59 barg in the model) — far above the header rating; flange or vessel failure, large gas release on the module, fire or explosion. With PSHH-001 also failed the model header peaks at 36.5 barg with PSV-001 relieving.	5×2 = 10	R01, R03
N2.1	Level MORE: Liquid carried into section 1	V-101 fills (with no trip the model reaches 100 % 82 s after the largest slug starts) and liquid enters section 1: impeller erosion or breakage, rotor imbalance, seal failure, gas release.	4×3 = 12	R04
N3.1	Flow LESS: Section 1 driven into surge	Surge: forward flow collapses and reverses every few seconds, thrust reverses, temperature builds up in the loop; seals and thrust bearing fail, rotor rub, gas release at the seals.	4×3 = 12	R11, R12, R38
N3.2	Measurement OTHER THAN: Anti-surge flow reads wrong	UIC-103 believes the margin is larger than it is: no recycle until the machine is already in surge on the next turndown.	4×3 = 12	R09, R12
N3.3	Control OTHER THAN: Anti-surge controller in MAN	No surge protection on turndown. With the wells cutting back and both loops in MAN the model surges 42 times in 5 min and TI-104 reaches 171 °C (no-trip run).	4×3 = 12	R10
N5.2	Flow LESS: Section 2 driven into surge	Surge in the denser HP section: violent flow reversals, thrust reversal, seal and bearing failure, gas release at 60–120 barg.	4×3 = 12	R09, R10, R12
N6.1	Flow NO: Discharge blocked at full load	The 30 m³ discharge header packs at full GT power: with no trip the model passes 138 barg 11 s after the closure and reaches 178 barg in 60 s — beyond the discharge design; rupture and a high-pressure gas release, fire or explosion.	5×2 = 10	R13, R14, R15
N6.7	Integrity OTHER THAN: Leak at 120 barg	High-pressure gas jet on the module: jet fire, explosion, escalation to the riser.	5×2 = 10	R37
N7.1	Pressure MORE: Line-pack rises	Pipeline and discharge pressure climb until PSHH-112 trips K-100; with no trip the model reaches 139 barg in 30 min — above the PSV-112 setting and possibly the pipeline MAOP: rupture of the line or riser.	4×3 = 12	R16, R17
N7.3	Integrity LESS: Pipeline or riser rupture	Massive subsea or riser gas release: a gas plume at the sea surface or a riser fire under the platform; the platform has no isolation between the riser and the line-pack except ESDV-113.	5×2 = 10	R18

Ref	Deviation	Consequence (no credit for safeguards)	S×L	Rec.
N8.1	Power NO: Loss of the driver	K-100 coasts down (at rest after about 7 min in the model); surge during the rundown unless both recycles open; the whole production (222 t/h) goes to flare until the wells are cut back; export lost.	3×4 = 12	R35
N9.2	Pressure NO: No lube oil	PSLL-150 trips K-100, but the train needs about 7 min to stop with no oil: bearings run dry through the coast-down (model bearing temperature 126 °C after 2 min with no trip).	4×3 = 12	R25
N10.1	Flow NO: Cooling medium lost	No cooling in either cooler: section 2 suction and export temperatures climb within a minute; with no trip the model reaches TI-109 205 °C and export 202 °C — seal failure and pipeline coating damage.	4×3 = 12	R27
N11.1	Flow NO: Blowdown fails	The loop stays at the settle-out pressure (≈ 69 barg) during a fire: vessel or pipe ruptures when the steel weakens; escalation.	5×2 = 10	R31, R13

5.2 Recommendations

No.	Recommendation	Node	Raised from	Highest risk	Action by
R01	Relief and flare study: confirm PSV-001 for the blocked-outlet case with the full production (K-100 tripped and PV-102 shut), and the flare header and tip for PV-102 at full production plus simultaneous blowdown (back-pressure on PSV-001 and the BDVs, radiation on the deck).	N1, N11	N1.4, N11.4	10 High	Process / Safety
R02	Specify the seat-leak class of PV-102 and test it at each shutdown; report flared gas as a loss figure (valued at the training gas price).	N1	N1.5	6 Medium	Mechanical / Operations
R03	Give PSHH-001 its own pressure transmitter: today PT-001 serves the PI-001 indication, PIC-102 and PSHH-001.	N1	N1.4	10 High	Instrument
R04	Slug handling: check LV-101 and the closed drain against the largest credible slug and agree a slug warning from production (well tests, pigging, restarts).	N2	N2.1	12 High	Process / Operations
R05	Gas blow-by protection of the closed drain: low-low level on V-101 and V-102 that closes LV-101 / LV-107, and relief of the drain drum sized for gas blow-by through a wide-open LV.	N2, N4	N2.2, N4.4	8 Medium	Process
R06	Put PSLL-101 on its own transmitter: the trip reads the PIC-101 measurement, so a frozen or drifting PIC-101 transmitter blinds the control loop, the alarm and the trip together.	N2	N2.3	9 Medium	Instrument
R07	Position switches on HV-101 and HV-102 with an alarm when either is open while the unit is in RUN.	N2, N11	N2.7, N11.6	9 Medium	Instrument
R08	Positive isolation for work on the compressor: double block-and-bleed or spades at the ESDVs; ESDV-101 and ESDV-113 leak tests at each proof test.	N12	N12.10	8 Medium	Operations / Mechanical
R09	Validate the anti-surge flow measurements: second flow element or a mass-balance deviation alarm, because UIC-103 and UIC-108 each act on one transmitter.	N3, N5	N3.2, N5.2	12 High	Instrument
R10	Alarm when UIC-103 or UIC-108 is not in AUTO with the unit in RUN; manual mode by supervisor access only.	N3, N5, N12	N3.3, N5.2, N12.3	12 High	Instrument / Operations
R11	Partial-stroke and stroke-time tests of UV-103 and UV-108; valve-position deviation alarm (ZI against controller output).	N3, N5	N3.1, N3.4, N5.1, N5.6	12 High	Instrument
R12	Independent surge detection: surge counter that trips I-1 on repeated surge cycles (for example 3 in 30 s), with the cycles logged for the rotating-equipment engineer.	N1, N2, N3, N5	N1.2, N2.6, N3.1, N3.2, N5.2	12 High	Instrument / Rotating
R13	Confirm and install PSV-107 (80 barg, V-102) and PSV-112 (138 barg, discharge), on hold on KC-PID-100: size for settle-out with fire, and for a blocked outlet at full GT power.	N2, N4, N6, N11	N2.4, N4.5, N6.1, N11.1	10 High	Process
R14	Review PSHH-112: set point and delay against PSV-112 (the dynamic check overshoots past the trip setting before the machine unloads); consider a shorter delay or a lower setting.	N6	N6.1	10 High	Instrument / Process
R15	ESDV-113: reduce spurious closure (redundant solenoids), proof-test closure time and seat leakage; inspect the NRV at each shutdown.	N6	N6.1, N6.5	10 High	Instrument / Mechanical
R16	Confirm the export pipeline MAOP against the highest discharge pressure (PSV-112 set point); if lower, add pipeline overpressure protection (HIPPS).	N7	N7.1	12 High	Pipeline

No.	Recommendation	Node	Raised from	Highest risk	Action by
R17	Agreement with the onshore terminal on intake restrictions with advance notice; bring PI-900 into the DCS with a high alarm.	N6, N7	N6.2, N7.1, N7.2	12 High	Operations / Instrument
R18	Riser ESD valve or subsea isolation valve near the platform and pipeline leak detection (pressure/flow balance with the terminal).	N7	N7.3	10 High	Pipeline / Safety
R19	Export gas quality: hydrocarbon dew-point limit on TIC-111 and the export gas, and a response plan for an AI-115 high alarm (new wells).	N1, N6, N7	N1.7, N6.4, N7.5	6 Medium	Process / Operations
R20	Corrosion management for wet CO_2 (12 mol %): water dew-point monitoring at the inlet, inhibitor, corrosion coupons in the header and the pipeline.	N1, N7	N1.8, N7.6	8 Medium	Materials / Process
R21	Fuel-gas conditioning (knock-out, superheater, filter — note 5) with a high-level trip, and a fuel source for start-up and low interstage pressure.	N4, N8	N4.3, N4.6, N8.5	8 Medium	Process
R22	Operating envelope for the operators: production that K-100 can take against ambient temperature and fuel-gas pressure (GT available power).	N1, N8	N1.3, N8.2	8 Medium	Operations / Process
R23	Overspeed protection: confirm the speed probes and voting of SSHH-100 and proof-test it with the GT supplier.	N8	N8.3	5 Medium	Instrument / Rotating
R24	Bearing protection: high-high bearing temperature trip or condition monitoring on TI-151 (alarm only today).	N3, N5, N9	N3.7, N5.4, N9.4	8 Medium	Rotating
R25	Lube oil: monitor the readiness of the standby pump (auto-start armed, pump healthy) with a DCS alarm; rundown tank (or DC emergency pump) that lubricates the bearings for the whole coast-down.	N9	N9.1, N9.2, N9.3	12 High	Rotating / Electrical
R26	Extend the HAZOP to the seal-gas panel and the lube-oil console sheets: seal-gas supply at start-up and at settle-out, primary-vent high-flow trip, gas in the oil reservoir.	N9	N9.5, N9.6, N9.7	8 Medium	Process / Rotating
R27	Installed spare cooling-medium pump with auto-start on P-160 failure.	N10	N10.1	12 High	Process / Mechanical
R28	Second temperature element on the cooler outlets (deviation alarm) so a single transmitter cannot move the cooling medium unseen.	N4, N6, N10	N4.1, N6.3, N10.3	9 Medium	Instrument
R29	Cooler performance monitoring (E-101/E-102 duty and approach, TV positions) and a cleaning programme for the coolers and the seawater exchanger.	N3, N4, N5, N10	N3.6, N4.1, N5.3, N5.5, N10.2	9 Medium	Operations
R30	Tube-rupture case for E-101 and E-102 (API 521): relief on the cooling-medium side and gas detection at the cooling-medium expansion tank.	N10	N10.5, N10.6	8 Medium	Process
R31	BDV-101/BDV-112 proof test (solenoid, stroke, restriction orifice) and confirmation of the 15-min blowdown with the fire case.	N11	N11.1	10 High	Instrument / Safety
R32	Low-temperature materials: confirm flare lines, BDV/HV-102 outlets and HV-101 for the expansion cooling (MDMT −46 °C, note 3, TBC).	N2, N11	N2.8, N11.3	8 Medium	Materials
R33	Flare knock-out drum, flare pilot monitoring and liquid carry-over protection (outside this node set; confirm in the flare study).	N11	N11.7, N11.8	8 Medium	Process
R34	ESD-2 criteria in the emergency procedure (when to blow down) and automatic ESD-2 from confirmed fire detection.	N11, N12	N11.5, N12.9	8 Medium	Operations / Safety
R35	Automatic production cut-back signal to the wells on a K-100 trip, and ramp-rate limits for production changes agreed with production control.	N7, N8, N12	N7.2, N8.1, N12.7, N12.8	12 High	Operations / Process
R36	Restart procedure: vent the settled-out loop with HV-102 before HV-101 or ESDV-101 is opened; simulator training on restart and loading.	N2, N12	N2.5, N12.3, N12.5	9 Medium	Operations
R37	Cover gas release, fire and gas detection and the ESD response of the compressor module in the fire-and-gas study (not on KC-PID-100).	N1, N6, N8, N9	N1.9, N6.7, N8.7, N9.6	10 High	Safety
R38	Feed-forward of the production flow FI-001 to UIC-103/UIC-108 (or rate-of-change limits) so a fast production cut does not drive the margins below the fast-opening line.	N1, N3	N1.1, N1.2, N3.1	12 High	Instrument / Process

5.3 Dynamic check of selected deviations on the simulator

A conventional HAZOP judges consequences qualitatively. Here ten deviations are run on copies of the K-100 model from the design state (200 MMSCFD, PIC-101 AUTO 30 barg in cascade to SIC-100, UIC-103/108 AUTO, PIC-102 AUTO 32 barg, TIC-106/111 and LIC-101/107 AUTO) with the existing safeguards active — alarms, the I-1 trips with their delays, PSHH-001, PSV-001, the P-150B auto-start, controllers in their normal modes — and **no operator action**, for 30 min. The result shows whether the safeguards act before harm and how much time the operator has. Alarms are checked every second with the page's alarm table and its suppression rules plus the DCS annunciations (trip, surge, power limit, pump stops, ESDV-113 shut, XV-001 shut); alarms already standing at design are not counted. The live simulator is not touched.

Run settings: kcRun from the design state (200 MMSCFD; PIC-101 AUTO 30 barg → SIC-100 CAS; UIC-103/108, PIC-102 32 barg, TIC-106/111, LIC-101/107 AUTO), deviation at t = 0, 1800 s, dt 0.25 s, SIS active, no operator action; alarms evaluated every 1 s with ALM + evalAlarms suppression.

Ref	Node	Deviation	Initiating event (on the model)	First alarms (mm:ss)	Trips and protective actions	Key tags: range in 30 min
D1	N8	Power NO (driver lost)	Gas turbine GT-100 flame-out at full load	• 00:00 K-100 TRIP • 00:01 FI-109 HI	GT-100 trip / flame-out at 00:00 train at rest at 07:10; loop settles out at 69.1 barg gas flared 109 t in 30 min (peak 243 t/h)	• SIC-100 speed 0.0 to 99.6 % • PI-001 header 30.30 to 32.00 barg • FI-109 flare 0 to 243 t/h
D2	N1	Flow LESS (production)	Wells cut back: production gas falls by 55 % within seconds	• 00:08 PIC-101 LO • 00:19 FI-114 LO • 00:23 UIC-103 LO • 00:23 UIC-108 LO • 00:27 FI-109 HI	no trip in 30 min (end: speed 80.0 %, PI-112 84.8 barg)	• SIC-100 speed 80.0 to 99.9 % • UIC-103 margin 4.8 to 60.8 % • PI-112 discharge 84.8 to 120.9 barg
D3	N10	Flow NO (cooling medium)	Cooling-medium pump P-160 stops; no cooling to E-101 or E-102	• 00:00 P-160 STOP • 00:28 TIC-106 HI • 00:41 TIC-111 HI • 00:47 TI-109 HI • 01:28 K-100 TRIP • 01:29 FI-109 HI	TSHH-109 section 2 discharge ≥ 150 °C at 01:28 train at rest at 09:04; loop settles out at 70.4 barg gas flared 103 t in 30 min (peak 256 t/h)	• TIC-106 (E-101 out) 40.0 to 101.0 °C • TI-109 58.5 to 152.6 °C • TIC-111 (export) 45.0 to 105.2 °C
D4	N5	Flow MORE (recycle)	HP anti-surge valve UV-108 goes to its fail-open position (air lost)	• 00:01 FI-114 LO • 00:04 FI-109 HI • 00:07 PIC-101 HI • 00:08 JI-100 POWER LIMIT • 00:12 PIC-102 HI • 01:03 SIC-100 HI • +2 more	no trip in 30 min (end: speed 105.0 %, PI-112 94.5 barg) gas flared 44 t in 30 min (peak 189 t/h)	• FI-114 export 0 to 180 MMSCFD • FI-109 flare 0 to 189 t/h • SIC-100 speed 99.9 to 105.5 %
D5	N6	Flow NO (export blocked)	Export ESD valve ESDV-113 closes spuriously with K-100 at full load	• 00:00 ESDV-113 SHUT • 00:01 FI-114 LO • 00:04 PI-112 HI • 00:10 FI-109 HI • 00:10 K-100 TRIP	PSHH-112 discharge ≥ 132 barg at 00:10 train at rest at 07:08; loop settles out at 73.1 barg gas flared 108 t in 30 min (peak 249 t/h)	• PI-112 discharge 72.8 to 135.5 barg • UIC-108 margin 15.2 to 100.0 % • FI-109 flare 0 to 249 t/h
D6	N9	Pressure LESS (lube oil)	Main lube-oil pump P-150A trips and standby P-150B fails to start	• 00:02 PI-150 LO • 00:05 K-100 TRIP • 00:06 FI-109 HI	PSLL-150 lube oil pressure ≤ 1.2 barg at 00:05 train at rest at 07:15; loop settles out at 69.1 barg gas flared 108 t in 30 min (peak 243 t/h)	• PI-150 lube oil 0.00 to 2.75 barg • SIC-100 speed 0.0 to 99.9 % • FI-109 flare 0 to 243 t/h
D7	N7	Pressure MORE (arrival)	Onshore terminal restricts intake: arrival pressure +30 bar at once	• 02:14 PI-112 HI • 06:40 K-100 TRIP • 06:41 FI-109 HI	PSHH-112 discharge ≥ 132 barg at 06:40 train at rest at 13:39; loop settles out at 73.5 barg gas flared 84 t in 30 min (peak 247 t/h)	• PI-112 discharge 73.2 to 132.0 barg • SIC-100 speed 0.0 to 102.0 % • JI-100 shaft power 0.00 to 18.21 MW
D8	N2	Level MORE (liquid slug)	Liquid slug from the production header: 12 kg/s of liquid into V-101 for 2 min	• 00:34 LIC-101 HI • 01:05 FI-109 HI • 01:05 K-100 TRIP	LSHH-101 V-101 liquid level high-high at 01:05 train at rest at 08:14; loop settles out at 69.1 barg gas flared 105 t in 30 min (peak 243 t/h)	• LIC-101 25 to 87 % • SIC-100 speed 0.0 to 99.9 % • FI-109 flare 0 to 243 t/h
D9	N3	Control OTHER THAN (anti-surge in MAN)	UIC-103 left in MAN with UV-103 shut while the wells cut back by 55 %	• 00:08 PIC-101 LO • 00:19 FI-114 LO • 00:21 UIC-103 LO • 00:22 UIC-108 LO • 00:22 VI-105 HI • 00:22 UIC-103 SURGE • +5 more	TSHH-104 section 1 discharge ≥ 150 °C at 00:54 surge cycles: section 1 9, section 2 1 (first at 00:22) train at rest at 06:09; loop settles out at 67.2 barg gas flared 47 t in 30 min (peak 190 t/h)	• UIC-103 margin −20.0 to 100.0 % • SIC-100 speed 0.0 to 99.9 % • TI-104 34.9 to 156.0 °C

Ref	Node	Deviation	Initiating event (on the model)	First alarms (mm:ss)	Trips and protective actions	Key tags: range in 30 min
D10	N1	Pressure MORE (header, flare valve shut)	GT-100 flame-out while PIC-102 is in MAN at 0 % (PV-102 does not open on the trip)	• 00:00 K-100 TRIP • 00:04 PIC-102 HI • 00:11 XV-001 SHUT	GT-100 trip / flame-out at 00:00 XV-001 shuts the wells at 00:11 PSV-001 lifts at 00:11 (up to 3.2 t/h) train at rest at 07:10; loop settles out at 69.2 barg	• PI-001 header 30.30 to 36.00 barg • FI-109 flare 0 to 1 t/h • SIC-100 speed 0.0 to 99.6 %

What the runs tell the HAZOP team

• **Driver lost (N8):** the trip is immediate and both recycles open: no surge cycle in the coast-down (at rest at 07:10). PV-102 holds the header at 32.00 barg but flares up to 243 t/h — 109 t of gas in 30 min. The process is safe; the loss is the flare (R35).

• **Production falls 55 % (N1):** no trip; the section 1 margin dips to 4.8 % — below the 10 % control line, into the fast-opening band (5 %) — before the speed comes down to 80.0 %. The anti-surge loops are the only thing between a fast production cut and surge (R38).

• **Cooling medium lost (N10):** P-160 STOP at once, TIC-106 HI at 00:28, TI-109 HI at 00:47 and TSHH-109 trips K-100 at 01:28. About a minute and a half from pump stop to trip: the operator cannot restore cooling in time, so a spare with auto-start is the practical layer (R27).

• **UV-108 fails open (N5):** no trip at all. Export falls to 0 MMSCFD at first, GT-100 hits its power limit in seconds, the speed reaches 105.5 % (below SSHH-100 at 110 %) and PV-102 flares up to 189 t/h — 44 t in 30 min. An operability and loss case the alarms announce clearly (R11).

• **Export blocked by ESDV-113 (N6):** PI-112 HI at 00:04, PSHH-112 trips at 00:10, yet the discharge still overshoots to 135.5 barg — only a few bar below the 138-barg PSV-112 that is on hold on the P&ID. Without the trip the model passes 138 barg in about 11 s (R13, R14).

• **Lube oil lost with a failed standby (N9):** PI-150 LO at 00:02 and PSLL-150 trips at 00:05 — the trip is fast, but the train turns until 07:15 with no oil pressure. The SIF cannot protect the bearings during the coast-down; a rundown tank can (R25).

• **Arrival pressure +30 bar (N7):** PI-112 HI at 02:14 and PSHH-112 at 06:40: about 4 min for the operator to call the terminal or cut production. The terminal's pressure is visible only as telemetry, with no alarm (R17).

• **Liquid slug (N2):** LIC-101 HI at 00:34 and LSHH-101 trips at 01:05 with the level at 87 %: about 31 s between alarm and trip — no time for an operator. LSHH-101 is the real protection of section 1 (R04).

• **Anti-surge in MAN during a production cut (N3):** section 1 surges from 00:22 (9 cycles, 1 in section 2); vibration HI follows, and TSHH-104 finally trips at 00:54 with TI-104 at 156.0 °C. The trip ends the surge only indirectly, through the heat it creates (R10, R12).

• **Flare valve shut when K-100 trips (N1):** the header climbs to 36.00 barg; PSHH-001 shuts the wells at 00:11 and PSV-001 lifts at 00:11. Both last layers act, but PSHH-001 reads the same transmitter as PIC-102, so one failed transmitter defeats the flare control and the shut-in together (R01, R03).

5.4 Follow-up and close-out

• Each recommendation gets an owner and a due date; the owner answers *accepted and done*, *accepted with an alternative* (with justification) or *rejected* (with a justification agreed by the HAZOP leader and the asset manager). The register in section 8 is the tracking list.

• High scenarios whose protection relies on instrumented trips — the blocked discharge (PSHH-112, with PSV-112 on hold), liquid into section 1 (LSHH-101), surge with the anti-surge loop defeated (TSHH-104/VSHH-105 act only indirectly), header overpressure when PV-102 does not open (PSHH-001 sharing PT-001 with PIC-102), lube oil lost (PSLL-150) — go to **LOPA** to decide whether the trips are enough or a new SIF (surge counter, independent PSLL-101 and PSHH-001 transmitters, HIPPS on the pipeline) is needed (IEC 61511). See the Safety layers and LOPA tabs.

• Installing PSV-107 and PSV-112 (KC-PID-100 note 3, on hold), a spare cooling-medium pump or a lube-oil rundown tank is a change and goes through **management of change**, which checks whether this HAZOP must be updated.

• Extend the study to the drawings outside this scope (seal-gas panel, lube-oil console, cooling-medium and flare systems, fire and gas) and revalidate the HAZOP periodically (typically every five years) and after incidents or near misses.

• Use the simulator to train operators on the scenarios this HAZOP found — surge on turndown, settle-out and restart, the time left between a high alarm and a trip — and to time their response.

Appendix 1 · HAZOP worksheets

Node N1 · Production header

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Node**	N1 · Production header, PV-102 / PSV-001 to flare
Design intent	Receive dehydrated production gas (61.75 kg/s = 200 MMSCFD at 35 °C) from the HP separator through XV-001 at about 30.3 barg and pass it to K-100 through ESDV-101; keep the header at or below 32 barg by flaring the excess through PV-102 whenever K-100 cannot take it.	**Design conditions**	Production 200.0 MMSCFD (61.75 kg/s) at 30.30 barg, 35 °C; PV-102 closed, flare 0.0 t/h; wells shut-in pressure 59 barg (model); MW 22.27
Equipment	Production header DN500-PG-101 (150 m³ lumped), XV-001 well shut-in valve, PV-102 to flare, PSV-001, flare header FL-100	**Instruments**	FI-001, PI-001 / PT-001, PIC-102, PSHH-001, FI-109
Drawing	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Document**	KC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.1	Flow	**NO**	No production gas	• Wells shut in by production control or by XV-001 (PSHH-001, ESD-2) • HP separator trip upstream	Suction pressure falls while K-100 runs: the machine slows to minimum speed and recycles; below the PSLL-101 setting it trips; export stops and the pipeline line-pack falls (sales lost).	• PIC-101 LO 28.5 / HI 31.5 barg • UIC-103 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-103 cooled recycle, fail open • PSLL-101 at 20 barg for 5 s → I-1 (armed in RUN)	2	3	**6 M**	R38	Instrument / Process
N1.2	Flow	**LESS**	Production falls fast	• Wells cut back (fault library: production falls by up to 55 %) • well work • choke change upstream	K-100 turns down; both operating points move toward surge. If the anti-surge loops do not keep up the sections surge: flow reversal, thrust reversal, seal and bearing damage, possible gas release at the seals.	• UIC-103 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-103 cooled recycle, fail open • UIC-108 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-108 cooled recycle, fail open • UIC-103 margin LO 7 % and SURGE alarm • SURGE alarms UIC-103 / UIC-108 (flow reversal) • VSHH-105 at 90 µm for 3 s → I-1	4	2	**8 M**	R38, R12	Instrument / Process · Instrument / Rotating
N1.3	Flow	**MORE**	Production above the design rate	• New well on line (fault library: production rises by up to 15 %) • well test diverted to the header	GT-100 reaches its power limit; suction pressure rises and PV-102 flares the excess; the discharge and the pipeline pack toward PSHH-112.	• GT power-limit alarm (FV-100 ≥ 98 %) • PIC-101 LO 28.5 / HI 31.5 barg • PIC-102 (SP 32 barg) → PV-102 to flare, fail open • PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1	2	4	**8 M**	R22	Operations / Process
N1.4	Pressure	**MORE**	Header overpressure when K-100 stops	K-100 trips (any I-1 cause) with PV-102 failing to open: PIC-102 left in MAN at a low output, PV-102 stuck, flare line blocked	Full production into a closed 150 m³ header: the pressure climbs toward the wells' shut-in pressure (59 barg in the model) — far above the header rating; flange or vessel failure, large gas release on the module, fire or explosion. With PSHH-001 also failed the model header peaks at 36.5 barg with PSV-001 relieving.	• PIC-102 HI 32.5 barg • Trip feed-forward: on I-1 PIC-102 output goes to ≥ 80 % (production flared at once) • PSHH-001 at 34.5 barg for 3 s → XV-001 shuts the wells • PSV-001 set 36 barg on the production header → flare	5	2	**10 H**	R01, R03	Instrument · Process / Safety

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.5	Pressure	**LESS**	Header pressure falls: gas to flare	• PV-102 fails open (loss of air, fail-open design) • PV-102 seat passing (fault library) • PIC-102 set point entered too low	Production gas burnt in the flare instead of exported (lost sales and CO_2); suction pressure falls, K-100 slows and can trip on PSLL-101.	• FI-109 flare flow HI 5 t/h • PIC-101 LO 28.5 / HI 31.5 barg • PSLL-101 at 20 barg for 5 s → I-1 (armed in RUN)	2	3	**6 M**	R02	Mechanical / Operations
N1.6	Temperature	**MORE**	Production gas hotter than design	HP separator or upstream cooler upset	Hotter suction: more polytropic head per bar, higher discharge temperatures and shaft power; toward the TSHH-104 setting and the GT limit.	• TI-101 indication • TI-104 HI 125 °C • TSHH-104 at 150 °C for 3 s → I-1 • GT power-limit alarm (FV-100 ≥ 98 %)	2	2	**4 L**	—	—
N1.7	Composition	**OTHER THAN**	Richer (heavier) gas	New well with heavier gas (fault library: MW 22.3 → 24.7)	Lower speed of sound and more condensate in the coolers: higher V-102 level, liquids toward the fuel gas; the export gas can leave its hydrocarbon dew-point specification; volumetric meters read low.	• AI-115 molar mass HI 23.5 kg/kmol • LIC-107 → LV-107 to closed drain (fail closed) • LIC-107 LO 20 / HI 65 % • LSHH-107 at 85 % for 3 s → I-1	2	3	**6 M**	R19	Process / Operations
N1.8	Composition	**AS WELL AS**	Free water with CO_2-rich gas	• Dehydration upset upstream (outside the scope) • water carried from the HP separator	Wet CO_2 (12 mol %) corrodes carbon-steel header, coolers and pipeline; hydrates where the gas expands (HV-101, BDVs, PV-102).	• Upstream dehydration (outside the scope; assumed) • LIC-101 → LV-101 to closed drain (fail closed) • LIC-101 LO 20 / HI 65 %	3	2	**6 M**	R20	Materials / Process
N1.9	Integrity	**OTHER THAN**	Leak from the header	Flange or small-bore failure, PSV-001 inlet, corrosion	Gas release at about 30 barg on the compressor module: jet fire or explosion if ignited.	• Fire and gas detection and ESD (not on KC-PID-100; assumed) • Operator ESD-1 / ESD-2 on the SIS screen	4	2	**8 M**	R37	Safety

Node N2 · Inlet scrubber V-101

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Node**	N2 · Inlet scrubber V-101, ESDV-101 and HV-101
Design intent	Isolate the compressor loop at ESDV-101 (pressurise through HV-101 for start-up), remove liquids from the suction gas in V-101 (LIC-101 40 %, LV-101 fail closed to the closed drain) and deliver dry gas to section 1 at 30 barg (PIC-101).	**Design conditions**	Suction 30.00 barg, 34.4 °C; level 40 %, liquids 0.22 kg/s to the closed drain; settle-out after a trip ≈ 70.4 barg (Process-tab estimate)
Equipment	ESDV-101 with HV-101 bypass, V-101 inlet scrubber (25 m³ gas, mesh), LV-101 to closed drain, BDV-101 and HV-102 connections	**Instruments**	PIC-101, PSLL-101, TI-101, LIC-101, LSHH-101, HIC-101
Drawing	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Document**	KC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N2.1	Level	**MORE**	Liquid carried into section 1	• Liquid slug from production (fault library) • LV-101 stuck shut or the closed drain blocked • LIC-101 left in MAN	V-101 fills (with no trip the model reaches 100 % 82 s after the largest slug starts) and liquid enters section 1: impeller erosion or breakage, rotor imbalance, seal failure, gas release.	• LIC-101 → LV-101 to closed drain (fail closed) • LIC-101 LO 20 / HI 65 % • LSHH-101 at 85 % for 3 s → I-1	4	3	**12 H**	R04	Process / Operations
N2.2	Level	**LESS**	Gas blow-by to the closed drain	• LV-101 fails open or LIC-101 output high • level transmitter reads high	V-101 empties and gas at 30 barg flows into the low-pressure closed drain: drain drum overpressure, gas release at the drain vent.	• LIC-101 → LV-101 to closed drain (fail closed) • LIC-101 LO 20 / HI 65 % • Closed-drain system (separate sheet; assumed)	4	2	**8 M**	R05	Process
N2.3	Pressure	**LESS**	Suction pressure low, unseen	• PIC-101 transmitter freezes or drifts (fault library) • ESDV-101 partly shut	Speed control acts on a false pressure: the real suction can fall without alarm, the ratio and the discharge temperatures rise, the margins shrink. PSLL-101 reads the same measurement, so the trip is blind too.	• PIC-101 LO 28.5 / HI 31.5 barg (same transmitter) • PSLL-101 at 20 barg for 5 s → I-1 (armed in RUN) (same transmitter) • TSHH-104 at 150 °C for 3 s → I-1	3	3	**9 M**	R06	Instrument
N2.4	Pressure	**MORE**	Suction pressure high	• K-100 at its power or speed limit • PIC-101 in MAN • settle-out after a trip	Production is flared through PV-102; after a trip the whole loop settles out at ≈ 69 barg in V-101 (model), more than twice the header pressure.	• PIC-101 LO 28.5 / HI 31.5 barg • PIC-102 (SP 32 barg) → PV-102 to flare, fail open • Suction side designed for 80 barg (KC-PID-100 note 3, TBC), above the settle-out pressure	2	3	**6 M**	R13	Process
N2.5	Flow	**REVERSE**	Settled-out gas back into the header	ESDV-101 or HV-101 opened after a trip before the loop is vented	Gas at the settle-out pressure (≈ 69 barg) flows back into the 30-barg header: header overpressure, PSV-001 lifts.	• ESDV-101 open permissive: ΔP across the valve < 3 bar (pressurise with HV-101 first) • PSHH-001 at 34.5 barg for 3 s → XV-001 shuts the wells • PSV-001 set 36 barg on the production header → flare • Written procedures (Exercises tab: cold start-up, normal shutdown, ESD with blowdown, GT trip restart)	3	2	**6 M**	R36	Operations

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N2.6	Flow	**NO**	Suction isolated while running	ESDV-101 closes spuriously (solenoid, ESD signal fault)	Suction pressure collapses within seconds; the sections are driven toward surge during the trip and the rundown.	• PSLL-101 at 20 barg for 5 s → I-1 (armed in RUN) • UIC-103 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-103 cooled recycle, fail open • On I-1 UV-103 and UV-108 go to 100 % (trip opening)	3	2	**6 M**	R12	Instrument / Rotating
N2.7	Flow	**AS WELL AS**	Flow through the bypass after isolation	HV-101 left open after start-up	ESDV-101 shuts on a trip but gas keeps entering the loop through HV-101: isolation and blowdown are defeated (the BDV fights the inflow).	• HIC-101 output on the DCS • Written procedures (Exercises tab: cold start-up, normal shutdown, ESD with blowdown, GT trip restart)	3	3	**9 M**	R07	Instrument
N2.8	Temperature	**LESS**	Cold gas when pressurising	Pressurising a depressurised loop quickly through HV-101	Joule–Thomson cooling across HV-101 of about 13 K (0.45 K/bar from about 2 to 30 barg): hydrates in HV-101 and V-101, low-temperature embrittlement of the valve body.	Written procedures (Exercises tab: cold start-up, normal shutdown, ESD with blowdown, GT trip restart)	2	3	**6 M**	R32	Materials

Node N3 · Section 1 (LP) and UV-103 recycle

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Node**	N3 · Section 1 (LP) and UV-103 recycle
Design intent	Compress 61.75 kg/s from 30 to about 61 barg (r ≈ 2.0, Ep 0.685) at about 100 % speed, and keep the operating point at least 10 % from the surge line by recycling cooled gas from the V-102 outlet through UV-103 (fail open).	**Design conditions**	Q 2.116 m³/s at suction, q 0.999, r 1.997, Ep 0.685, discharge 60.9 barg / 105.2 °C, margin 61 %, UV-103 0 %, throat Mach 0.74
Equipment	K-100 section 1 (4 impellers), FT-103, UV-103 (DN300-PG-104 LP recycle)	**Instruments**	FI-103, UIC-103, PI-104, TI-104, TSHH-104, VI-105, VSHH-105
Drawing	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Document**	KC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N3.1	Flow	**LESS**	Section 1 driven into surge	• Fast production cut • UV-103 sticks (fault library) • anti-surge loop too slow or badly tuned	Surge: forward flow collapses and reverses every few seconds, thrust reverses, temperature builds up in the loop; seals and thrust bearing fail, rotor rub, gas release at the seals.	• UIC-103 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-103 cooled recycle, fail open • UIC-103 margin LO 7 % and SURGE alarm • VSHH-105 at 90 µm for 3 s → I-1 • TSHH-104 at 150 °C for 3 s → I-1	4	3	**12 H**	R11, R12, R38	Instrument · Instrument / Process · Instrument / Rotating
N3.2	Measurement	**OTHER THAN**	Anti-surge flow reads wrong	• FT-103 reads high (fault library) • impulse line plugged or leaking	UIC-103 believes the margin is larger than it is: no recycle until the machine is already in surge on the next turndown.	• SURGE alarms UIC-103 / UIC-108 (flow reversal) • VSHH-105 at 90 µm for 3 s → I-1	4	3	**12 H**	R09, R12	Instrument · Instrument / Rotating
N3.3	Control	**OTHER THAN**	Anti-surge controller in MAN	UIC-103 left in MAN after start-up, a valve stroke or tuning	No surge protection on turndown. With the wells cutting back and both loops in MAN the model surges 42 times in 5 min and TI-104 reaches 171 °C (no-trip run).	• SURGE alarms UIC-103 / UIC-108 (flow reversal) • UIC-103 margin LO 7 % and SURGE alarm • TSHH-104 at 150 °C for 3 s → I-1 • VSHH-105 at 90 µm for 3 s → I-1	4	3	**12 H**	R10	Instrument / Operations
N3.4	Flow	**NO**	No recycle when the trip demands it	UV-103 stuck at its running position (fault library) when I-1 calls for 100 %	The coast-down follows a falling speed with no recycle: surge cycles during the rundown, thrust-bearing and seal damage.	• On I-1 UV-103 and UV-108 go to 100 % (trip opening) (fails with the stuck valve) • SURGE alarms UIC-103 / UIC-108 (flow reversal)	3	2	**6 M**	R11	Instrument
N3.5	Flow	**MORE**	Stonewall (choke)	• UV-103 fails open or opens wide • low discharge pressure at start-up	Throat Mach approaches 1: head and efficiency collapse, vibration rises, the section cannot deliver.	• Stonewall alarm (impeller throat Mach ≥ 0.95) • VI-105 HI 60 µm	2	3	**6 M**	—	—
N3.6	Temperature	**MORE**	Section 1 discharge hot	• Section 1 fouling (fault library) • hot suction • high ratio • recycle at high flow	Discharge above the 150 °C sealing ceiling: seal and O-ring damage, gas leak at the shaft seals.	• TI-104 HI 125 °C • TSHH-104 at 150 °C for 3 s → I-1	3	3	**9 M**	R29	Operations
N3.7	Vibration	**MORE**	Rotor vibration high	Deposits, liquid carry-over, surge, misalignment	Bearing and seal damage, rub; a long run at high vibration ends in a seal failure and a gas release.	• VI-105 HI 60 µm • VSHH-105 at 90 µm for 3 s → I-1	3	2	**6 M**	R24	Rotating

Node N4 · Intercooler E-101

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Node**	N4 · Intercooler E-101, interstage scrubber V-102 and fuel-gas offtake
Design intent	Cool the section 1 discharge (about 106 °C) to 40 °C in E-101 (TIC-106 → TV-106, fail open), separate the condensate in V-102 (LIC-107 40 %, LV-107 fail closed), and supply section 2 at about 60.4 barg and GT-100 with fuel gas (1.38 kg/s).	**Design conditions**	E-101 duty 7.30 MW, TV-106 55 %, gas out 40.0 °C; V-102 60.41 barg, level 40 %, condensate 0.11 kg/s; fuel 1.38 kg/s
Equipment	E-101 intercooler (gas / cooling medium), V-102 interstage scrubber (40 m³ gas node with E-101), LV-107 to closed drain, fuel-gas offtake to the GT skid	**Instruments**	TIC-106, PI-107, LIC-107, LSHH-107
Drawing	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Document**	KC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N4.1	Temperature	**MORE**	Section 2 suction hot	• E-101 fouling (fault library) • TV-106 throttled by a false low reading (fault library: TT-106 reads low) or TIC-106 in MAN low • cooling medium hot or lost (node N10)	Hot gas into section 2: its discharge temperature, power and seal temperatures rise. With TV-106 shut and no trip the model's TI-109 passes 150 °C after 95 s and peaks at 193 °C.	• TIC-106 (SP 40 °C) → TV-106 fail open • TIC-106 LO 30 / HI 48 °C • TI-109 HI 130 °C • TSHH-109 at 150 °C for 3 s → I-1	3	3	**9 M**	R28, R29	Instrument · Operations
N4.2	Temperature	**LESS**	Interstage gas overcooled	• TV-106 fails open with cold cooling medium • TIC-106 set point too low	More hydrocarbon condensate in V-102; hydrates if water is present; higher V-102 level.	• TIC-106 (SP 40 °C) → TV-106 fail open • TIC-106 LO 30 / HI 48 °C • LIC-107 → LV-107 to closed drain (fail closed) • LIC-107 LO 20 / HI 65 % • LSHH-107 at 85 % for 3 s → I-1	2	3	**6 M**	—	—
N4.3	Level	**MORE**	Liquid into section 2 and the fuel gas	• LV-107 sticks shut (fault library) • richer gas (fault library) • overcooled interstage gas	Liquid carry-over into section 2 (impeller damage) and into the GT fuel gas (flame instability, hot streaks, flame-out).	• LIC-107 → LV-107 to closed drain (fail closed) • LIC-107 LO 20 / HI 65 % • LSHH-107 at 85 % for 3 s → I-1	4	2	**8 M**	R21	Process
N4.4	Level	**LESS**	Gas blow-by from V-102 to the closed drain	• LV-107 fails open • LIC-107 output high	Gas at 60 barg into the low-pressure closed drain: drain drum overpressure, gas release.	• LIC-107 → LV-107 to closed drain (fail closed) • LIC-107 LO 20 / HI 65 % • Closed-drain system (separate sheet; assumed)	4	2	**8 M**	R05	Process
N4.5	Pressure	**MORE**	Interstage at settle-out after a trip	Any I-1 trip: the loop equalises through the open recycle valves	E-101 and V-102 sit at the settle-out pressure (≈ 69 barg in the model) instead of 60 barg; a fire on the module would raise it further with no relief installed.	• Suction side designed for 80 barg (KC-PID-100 note 3, TBC), above the settle-out pressure • PSV-107 80 barg on V-102 (KC-PID-100 note 3: on hold, not drawn — not credited until installed) • ESD-2: BDV-101 and BDV-112 open to flare (fail open, RO 45 mm, choked), XV-001 shuts the wells	4	2	**8 M**	R13	Process

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N4.6	Pressure	**LESS**	Fuel gas pressure low	• Interstage pressure low during start-up, turndown or after a trip • fuel-gas skid restricted (fault library: fuel gas pressure low)	GT-100 loses available power or cannot accelerate; speed falls, suction rises and PV-102 flares; a flame-out trips the unit.	• GT power-limit alarm (FV-100 ≥ 98 %) • PIC-102 (SP 32 barg) → PV-102 to flare, fail open • GT-100 flame-out trips I-1 • XV-100 shuts the fuel	2	3	**6 M**	R21	Process

Node N5 · Section 2 (HP) and UV-108 recycle

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Node**	N5 · Section 2 (HP) and UV-108 recycle
Design intent	Compress 60.26 kg/s from about 60.4 to 121.6 barg (r ≈ 2.0, Ep 0.655) and keep at least 10 % surge margin by recycling cooled gas from after E-102 to before E-101 through UV-108 (fail open).	**Design conditions**	Q 0.998 m³/s at suction, q 1.001, r 1.997, Ep 0.655, discharge 121.6 barg / 115.7 °C, margin 61 %, UV-108 0 %
Equipment	K-100 section 2 (4 impellers), UV-108 (DN250-PG-107 HP recycle)	**Instruments**	FI-108, UIC-108, PI-109, TI-109, TSHH-109, VI-110, VSHH-110, TI-151
Drawing	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Document**	KC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N5.1	Flow	**MORE**	Large HP recycle	• UV-108 fails open (fault library: air or solenoid failure) • UIC-108 output high	Most of the section 2 flow recirculates: export falls sharply, GT-100 goes to its power limit, suction rises and PV-102 flares production; the machine runs at its top speed near stonewall.	• FI-114 export LO 150 MMSCFD • GT power-limit alarm (FV-100 ≥ 98 %) • PIC-102 (SP 32 barg) → PV-102 to flare, fail open • FI-109 flare flow HI 5 t/h • SSHH-100 at 110 % for 0.5 s → I-1	2	3	**6 M**	R11	Instrument
N5.2	Flow	**LESS**	Section 2 driven into surge	• Production cut • arrival pressure rising (fault library) • UIC-108 in MAN or slow	Surge in the denser HP section: violent flow reversals, thrust reversal, seal and bearing failure, gas release at 60–120 barg.	• UIC-108 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-108 cooled recycle, fail open • UIC-108 margin LO 7 % and SURGE alarm • VSHH-110 at 90 µm for 3 s → I-1 • TSHH-109 at 150 °C for 3 s → I-1	4	3	**12 H**	R09, R10, R12	Instrument · Instrument / Operations · Instrument / Rotating
N5.3	Temperature	**MORE**	Section 2 discharge hot	• Hot section 2 suction (node N4) • section 2 fouling (fault library) • high ratio from a high arrival pressure	Discharge above the 150 °C sealing ceiling: seal damage and gas leak.	• TI-109 HI 130 °C • TSHH-109 at 150 °C for 3 s → I-1	3	3	**9 M**	R29	Operations
N5.4	Vibration	**MORE**	HP-end vibration and bearing temperature rise	• Section 2 bearing wear (fault library) • imbalance • surge	Bearing failure, rub, seal failure and gas release; long repair.	• VI-110 HI 60 µm • VSHH-110 at 90 µm for 3 s → I-1 • TI-151 bearing HI 95 °C	4	2	**8 M**	R24	Rotating
N5.5	Pressure	**MORE**	Section ratio split shifts	Section 1 or section 2 fouling (fault library)	One section takes more ratio: hotter discharge, more power, smaller margin in that section.	• TI-104 HI 125 °C • TI-109 HI 130 °C • SIC-100 LO 72 / HI 104 %	2	3	**6 M**	R29	Operations
N5.6	Timing	**LATE**	Recycle opens too slowly on a trip	Slow actuator, undersized solenoid or exhaust on UV-108	Section 2 surges in the first seconds of the coast-down (the speed falls faster than the recycle opens).	• On I-1 UV-103 and UV-108 go to 100 % (trip opening) • SURGE alarms UIC-103 / UIC-108 (flow reversal)	3	2	**6 M**	R11	Instrument

Node N6 · Aftercooler E-102

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Node**	N6 · Aftercooler E-102, discharge header and ESDV-113
Design intent	Cool the section 2 discharge (about 116 °C) to 45 °C in E-102 (TIC-111 → TV-111, fail open) and deliver export gas at about 121 barg through the NRV, ESDV-113 and FE-114 to the riser; PSHH-112 protects against a blocked outlet.	**Design conditions**	E-102 duty 7.73 MW, TV-111 49 %, export 45.0 °C at 120.9 barg; FI-114 195.2 MMSCFD
Equipment	E-102 aftercooler, discharge header DN400-EG-108 (30 m³), NRV, ESDV-113, FE-114, BDV-112 connection	**Instruments**	TIC-111, PI-112, PSHH-112, FI-114, PI-114, AI-115
Drawing	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Document**	KC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N6.1	Flow	**NO**	Discharge blocked at full load	• ESDV-113 closes spuriously (fault library: solenoid fault) • downstream block valve shut • NRV stuck shut	The 30 m³ discharge header packs at full GT power: with no trip the model passes 138 barg 11 s after the closure and reaches 178 barg in 60 s — beyond the discharge design; rupture and a high-pressure gas release, fire or explosion.	• ESDV-113 SHUT alarm with the compressor running • PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1 • PSV-112 138 barg on the discharge (KC-PID-100 note 3: on hold, not drawn — not credited until installed)	5	2	**10 H**	R13, R14, R15	Instrument / Mechanical · Instrument / Process · Process
N6.2	Pressure	**MORE**	Discharge pressure rising slowly	• Onshore arrival pressure rising (fault library) • production above design with the pipeline packing	K-100 works harder until PSHH-112 trips it: export lost and production flared until the wells are cut back.	• PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1 • PI-900 arrival pressure telemetry from the onshore terminal (indication only, no alarm)	3	3	**9 M**	R17	Operations / Instrument
N6.3	Temperature	**MORE**	Export gas hot	• TV-111 shut or TIC-111 in MAN low • cooling medium lost (N10) • E-102 fouling	Export gas above the pipeline coating limit (assumed 60 °C): coating damage, upheaval buckling of the line. With no cooling and no trip the model's export temperature reaches 202 °C.	• TIC-111 (SP 45 °C) → TV-111 fail open • TIC-111 LO 30 / HI 55 °C • TSHH-109 at 150 °C for 3 s → I-1	3	2	**6 M**	R28	Instrument
N6.4	Temperature	**LESS**	Export gas too cold	• TV-111 fails open with cold cooling medium • TIC-111 set point too low	Export gas below its hydrocarbon dew point: liquids in the pipeline, slugs at the onshore terminal.	• TIC-111 (SP 45 °C) → TV-111 fail open • TIC-111 LO 30 / HI 55 °C	2	2	**4 L**	R19	Process / Operations
N6.5	Flow	**REVERSE**	Pipeline gas back into the compressor	After a trip the NRV passes and ESDV-113 fails to close	Line-pack (450 m³ at about 120 barg) flows back through E-102 and section 2: reverse rotation, the loop rises toward pipeline pressure above the suction-side design, fire if the seals fail.	• Check valve (NRV) upstream of ESDV-113 • ESDV-113 shuts on I-1 and ESD • ESD-2: BDV-101 and BDV-112 open to flare (fail open, RO 45 mm, choked), XV-001 shuts the wells	4	2	**8 M**	R15	Instrument / Mechanical
N6.6	Pressure	**LESS**	Discharge pressure low	• Onshore demand high (fault library: arrival pressure falls) • pipeline leak (node N7)	Lower ratio, the machine slows; flow stays at the production rate (operability only, unless the cause is a leak).	• SIC-100 LO 72 / HI 104 % • FI-114 export LO 150 MMSCFD	1	3	**3 L**	—	—

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N6.7	Integrity	**OTHER THAN**	Leak at 120 barg	• Flange, FE-114 or small-bore failure • vibration fatigue	High-pressure gas jet on the module: jet fire, explosion, escalation to the riser.	• Fire and gas detection and ESD (not on KC-PID-100; assumed) • Operator ESD-1 / ESD-2 on the SIS screen • ESD-2: BDV-101 and BDV-112 open to flare (fail open, RO 45 mm, choked), XV-001 shuts the wells	5	2	**10 H**	R37	Safety

Node N7 · Export riser and pipeline to the onshore terminal

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Node**	N7 · Export riser and pipeline to the onshore terminal
Design intent	Transport about 60.26 kg/s (195 MMSCFD) at about 120 barg and 45 °C through the export riser and 36-in pipeline (lumped teaching volume 450 m³) to the onshore terminal at about 70 barg (PI-900), inside the gas specification (AI-115).	**Design conditions**	Inlet 120.1 barg, arrival 70.0 barg, 195.2 MMSCFD, gas velocity 0.9 m/s, MW 22.27
Equipment	Export riser, subsea pipeline (Weymouth-type line), onshore receiving terminal (outside the scope)	**Instruments**	FI-114, PI-114, PI-900 (telemetry), AI-115
Drawing	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Document**	KC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N7.1	Pressure	**MORE**	Line-pack rises	• Onshore terminal restricts its intake (fault library: arrival pressure rises by up to 30 bar) • terminal pressure control fault	Pipeline and discharge pressure climb until PSHH-112 trips K-100; with no trip the model reaches 139 barg in 30 min — above the PSV-112 setting and possibly the pipeline MAOP: rupture of the line or riser.	• PI-900 arrival pressure telemetry from the onshore terminal (indication only, no alarm) • PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1	4	3	**12 H**	R16, R17	Operations / Instrument · Pipeline
N7.2	Flow	**NO**	Onshore intake shut	• Terminal inlet ESD valve closes • terminal blackout	The pipeline packs at the full export rate; PSHH-112 trips K-100 and production is flared until the wells are cut back.	• PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1 • Trip feed-forward: on I-1 PIC-102 output goes to ≥ 80 % (production flared at once)	3	2	**6 M**	R17, R35	Operations / Instrument · Operations / Process
N7.3	Integrity	**LESS**	Pipeline or riser rupture	Anchor drag or dropped object, corrosion (wet CO_2), riser damage by a vessel	Massive subsea or riser gas release: a gas plume at the sea surface or a riser fire under the platform; the platform has no isolation between the riser and the line-pack except ESDV-113.	• ESDV-113 shuts on I-1 and ESD • PI-114 / FI-114 indications	5	2	**10 H**	R18	Pipeline / Safety
N7.4	Pressure	**LESS**	Arrival pressure low	Onshore demand high (fault library)	Export flow stays at the production rate; the machine unloads slightly (operability only).	PI-900 arrival pressure telemetry from the onshore terminal (indication only, no alarm)	1	3	**3 L**	—	—
N7.5	Composition	**OTHER THAN**	Export gas off specification	• Richer gas from a new well (fault library) • export too cold (node N6)	Hydrocarbon dew point exceeded: liquids in the pipeline, sales gas rejected at the terminal.	• AI-115 molar mass HI 23.5 kg/kmol • TIC-111 (SP 45 °C) → TV-111 fail open • TIC-111 LO 30 / HI 55 °C	2	3	**6 M**	R19	Process / Operations
N7.6	Composition	**AS WELL AS**	Water in the export gas	Dehydration upset upstream (outside the scope)	Internal corrosion of the pipeline by wet CO_2, hydrates during shut-ins.	Upstream dehydration (outside the scope; assumed)	4	2	**8 M**	R20	Materials / Process

Node N8 · GT-100 driver and fuel gas

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Node**	N8 · GT-100 driver and fuel gas
Design intent	Drive K-100 directly at 9500 rpm = 100 % (SIC-100 range 70–105 %) with up to 19.86 MW available at 30 °C (22 MW ISO, −0.65 %/°C above 15 °C), burning fuel gas from V-102 through XV-100 and FV-100 (fail closed).	**Design conditions**	Speed 99.9 %, shaft 16.69 MW of 19.86 MW available (margin 3.16 MW), FV-100 84 %, fuel 4.97 t/h, exhaust 574 °C, ambient 30 °C
Equipment	GT-100 two-shaft industrial gas turbine, fuel-gas skid, XV-100, FV-100, exhaust	**Instruments**	SIC-100, SSHH-100, JI-100, FI-100, TI-100, TI-099
Drawing	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Document**	KC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N8.1	Power	**NO**	Loss of the driver	• GT-100 flame-out (fault library) • fuel valve closes (fail closed on air loss) • GT internal protection	K-100 coasts down (at rest after about 7 min in the model); surge during the rundown unless both recycles open; the whole production (222 t/h) goes to flare until the wells are cut back; export lost.	• GT-100 flame-out trips I-1 • XV-100 shuts the fuel • On I-1 UV-103 and UV-108 go to 100 % (trip opening) • Trip feed-forward: on I-1 PIC-102 output goes to ≥ 80 % (production flared at once) • PSHH-001 at 34.5 barg for 3 s → XV-001 shuts the wells	3	4	**12 H**	R35	Operations / Process
N8.2	Power	**LESS**	GT available power falls	• Hot ambient (fault library) • fuel-gas pressure low (fault library) • GT compressor fouling	GT at its limit: speed sags, suction rises above set point and PV-102 flares; with extra production the shortfall grows.	• GT power-limit alarm (FV-100 ≥ 98 %) • SIC-100 LO 72 / HI 104 % • PIC-102 (SP 32 barg) → PV-102 to flare, fail open	2	4	**8 M**	R22	Operations / Process
N8.3	Speed	**MORE**	Overspeed	• Sudden loss of load (coupling failure, large recycle with the governor slow) • SIC-100 or FV-100 failure	Rotor and impeller burst at excessive speed: fragments through the casing, fatal injuries. Normal top speed is 105 %; with UV-108 failed open the model reaches 105.5 %.	• SIC-100 LO 72 / HI 104 % • SSHH-100 at 110 % for 0.5 s → I-1 • FV-100 fuel valve fail closed • XV-100 fuel ESD valve	5	1	**5 M**	R23	Instrument / Rotating
N8.4	Temperature	**MORE**	GT exhaust hot	Hot ambient, GT fouling, over-firing at the power limit	Hot-section damage (blades, combustor), loss of availability.	• TI-100 exhaust HI 650 °C • GT power-limit alarm (FV-100 ≥ 98 %)	3	2	**6 M**	—	—
N8.5	Composition	**AS WELL AS**	Liquids in the fuel gas	• V-102 carry-over (node N4) • richer gas condensing in the fuel line	Flame instability, hot streaks, combustor damage, flame-out (driver lost).	• LIC-107 → LV-107 to closed drain (fail closed) • LIC-107 LO 20 / HI 65 % • LSHH-107 at 85 % for 3 s → I-1	3	2	**6 M**	R21	Process
N8.6	Sequence	**PART OF**	Ignition without a complete purge	Purge cut short or skipped after a failed start	Fuel accumulated in the combustor and exhaust ignites: explosion in the GT enclosure or exhaust duct.	GT start permissives (SIS reset, PI-150 ≥ 2.0 barg, P-160 running, ESDV-101 open with suction ≥ 20 barg, UV-103/108 ≥ 90 %, scrubber levels < 60 %, BDVs closed) and the 30 s crank-purge before ignition	4	1	**4 L**	—	—
N8.7	Integrity	**OTHER THAN**	Fuel-gas leak in the GT enclosure	Fuel-line flange or fitting failure, vibration	Gas accumulates in the enclosure: explosion.	Fire and gas detection and ESD (not on KC-PID-100; assumed)	4	2	**8 M**	R37	Safety

Node N9 · Lube oil and seal system

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Node**	N9 · Lube oil and seal system
Design intent	Supply lube oil at about 3.0 barg to the K-100 and GT bearings (P-150A main, P-150B standby auto-starting at 1.8 barg with its selector in AUTO) while running and through the coast-down; keep the process gas inside the casing with the shaft seals (seal-gas panel on a separate sheet).	**Design conditions**	PI-150 3.00 barg, bearing 74.5 °C; trip at 1.2 barg, standby start at 1.8 barg
Equipment	Lube-oil console (P-150A/B, reservoir, coolers, filters), shaft seals and seal-gas panel (separate sheet)	**Instruments**	PI-150, PSLL-150, TI-151
Drawing	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Document**	KC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N9.1	Pressure	**LESS**	Main lube-oil pump stops	• P-150A trips (fault library) • motor or power fault	Without oil the bearings wipe within seconds at full speed: rotor damage, seal failure and gas release.	• PI-150 LO 1.8 barg • P-150B auto-starts at 1.8 barg (selector AUTO) • P-150B RUN alarm (main pump stopped) • PSLL-150 at 1.2 barg for 2 s → I-1	4	2	**8 M**	R25	Rotating / Electrical
N9.2	Pressure	**NO**	No lube oil	Main pump trips and the standby is not available on demand (fault library: standby fails to start)	PSLL-150 trips K-100, but the train needs about 7 min to stop with no oil: bearings run dry through the coast-down (model bearing temperature 126 °C after 2 min with no trip).	• PI-150 LO 1.8 barg • P-150B auto-starts at 1.8 barg (selector AUTO) • PSLL-150 at 1.2 barg for 2 s → I-1	4	3	**12 H**	R25	Rotating / Electrical
N9.3	Power	**NO**	Platform power failure	Main power lost: both AC lube-oil pumps stop together	As above: no lubrication during the coast-down.	PSLL-150 at 1.2 barg for 2 s → I-1	4	2	**8 M**	R25	Rotating / Electrical
N9.4	Temperature	**MORE**	Bearing temperature high	• Bearing wear (fault library) • lube-oil cooler fouled or cooling medium hot • oil degraded	Bearing failure, rub, seal failure.	• TI-151 bearing HI 95 °C • VSHH-110 at 90 µm for 3 s → I-1	3	2	**6 M**	R24	Rotating
N9.5	Composition	**AS WELL AS**	Gas or water in the lube oil	• Seal leakage into the oil • lube-oil cooler tube leak	Foaming, loss of oil film, bearing failure; flammable gas from the reservoir vent.	Oil sampling and reservoir vent (assumed)	3	2	**6 M**	R26	Process / Rotating
N9.6	Flow	**MORE**	Seal leakage high	Shaft-seal damage by surge thrust, liquid carry-over or discharge above 150 °C	Process gas to the seal vent or into the compressor house: fire or explosion.	• Seal-gas panel with its own alarms (separate sheet, not modelled; assumed) • TSHH-104 at 150 °C for 3 s → I-1 • TSHH-109 at 150 °C for 3 s → I-1	4	2	**8 M**	R26, R37	Process / Rotating · Safety
N9.7	Pressure	**LESS**	Seal-gas supply lost	• No discharge pressure at start-up or after settle-out • seal-gas filter blocked	Unfiltered process gas reaches the seal faces: seal damage and a later leak.	Seal-gas panel with its own alarms (separate sheet, not modelled; assumed)	3	2	**6 M**	R26	Process / Rotating

Node N10 · Cooling medium

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Node**	N10 · Cooling medium: P-160, TV-106, TV-111
Design intent	Circulate cooling medium at 30 °C (TI-160) with P-160 through E-101 and E-102 (TV-106 and TV-111 fail open, about half open at design) and reject 15.2 MW to seawater.	**Design conditions**	Supply 30.0 °C; TV-106 55 % (≈ 43 kg/s), TV-111 49 % (≈ 40 kg/s); duties 7.30 + 7.73 MW
Equipment	P-160, cooling-medium supply and return headers, TV-106, TV-111, seawater exchanger (separate sheet)	**Instruments**	TI-160, TIC-106, TIC-111
Drawing	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Document**	KC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N10.1	Flow	**NO**	Cooling medium lost	• P-160 trips (fault library) • header valve closed • power failure	No cooling in either cooler: section 2 suction and export temperatures climb within a minute; with no trip the model reaches TI-109 205 °C and export 202 °C — seal failure and pipeline coating damage.	• P-160 STOP alarm with the unit running • TIC-106 (SP 40 °C) → TV-106 fail open • TIC-106 LO 30 / HI 48 °C • TI-109 HI 130 °C • TSHH-109 at 150 °C for 3 s → I-1	4	3	**12 H**	R27	Process / Mechanical
N10.2	Temperature	**MORE**	Cooling medium supply hot	• Seawater exchanger fouled (fault library: cooling medium supply hot) • hot seawater	TV-106/TV-111 open fully and the coolers lose control: hotter section 2 and export, more power.	• TI-160 cooling-medium supply indication (no alarm) • TIC-106 (SP 40 °C) → TV-106 fail open • TIC-106 LO 30 / HI 48 °C • TIC-111 (SP 45 °C) → TV-111 fail open • TIC-111 LO 30 / HI 55 °C • TI-109 HI 130 °C	2	3	**6 M**	R29	Operations
N10.3	Flow	**LESS**	Cooling medium to one cooler throttled	• TIC-106 acting on a false low reading (fault library: TT-106 reads low) • TIC-106 or TIC-111 in MAN with a low output	The gas leaves that cooler hotter than shown; section 2 or the export runs hot (see N4, N6).	• TI-109 HI 130 °C • TSHH-109 at 150 °C for 3 s → I-1 • TIC-111 (SP 45 °C) → TV-111 fail open • TIC-111 LO 30 / HI 55 °C	3	3	**9 M**	R28	Instrument
N10.4	Flow	**MORE**	Coolers over-cooled	TV-106 and TV-111 fail open together (instrument-air failure)	More condensate in V-102, export closer to its dew point (operability).	• TIC-106 (SP 40 °C) → TV-106 fail open • TIC-106 LO 30 / HI 48 °C • LIC-107 → LV-107 to closed drain (fail closed) • LIC-107 LO 20 / HI 65 %	2	3	**6 M**	—	—
N10.5	Pressure	**MORE**	Gas into the cooling-medium circuit	Tube rupture in E-101 (60 barg) or E-102 (121 barg)	High-pressure gas enters the low-pressure cooling-medium circuit: exchanger shell and P-160 overpressure, gas released at the expansion tank.	Cooling-medium system design (separate sheet; assumed)	4	2	**8 M**	R30	Process
N10.6	Composition	**AS WELL AS**	Gas in the cooling-medium return	Small tube leak	Gas at the expansion tank vent, P-160 vapour-locked (then cooling lost).	Cooling-medium system design (separate sheet; assumed)	3	2	**6 M**	R30	Process

Node N11 · Blowdown BDV-101 / BDV-112

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Node**	N11 · Blowdown BDV-101 / BDV-112, HV-102 and flare
Design intent	On ESD-2 depressurise the compressor loop through BDV-101 (suction side) and BDV-112 (discharge), RO 45 mm and choked, to 6.9 barg within 15 min (API 521); after a shutdown vent the settled-out loop with HV-102; carry PV-102, PSV-001, BDV and HV-102 flows safely to the flare.	**Design conditions**	Loop inventory to blow down: S1 25 + D1 40 + D2 30 m³; settle-out ≈ 70.4 barg; model blowdown to 6.9 barg in 6.1 min, flare line down to −21 °C
Equipment	BDV-101, BDV-112 (fail open, RO 45 mm), HV-102, flare header DN500-FL-100, flare tip	**Instruments**	FI-109, HIC-102
Drawing	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Document**	KC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N11.1	Flow	**NO**	Blowdown fails	BDV stuck shut, solenoid fails energised, restriction orifice blocked by hydrate	The loop stays at the settle-out pressure (≈ 69 barg) during a fire: vessel or pipe ruptures when the steel weakens; escalation.	• ESD-2: BDV-101 and BDV-112 open to flare (fail open, RO 45 mm, choked), XV-001 shuts the wells • BDV OPEN alarm • PSV-107 80 barg on V-102 (KC-PID-100 note 3: on hold, not drawn — not credited until installed)	5	2	**10 H**	R31, R13	Instrument / Safety · Process
N11.2	Flow	**MORE**	Blowdown at full load	BDV opens spuriously with K-100 running	Large unplanned flaring; the unit is lost; low-temperature flow in the flare line.	• BDV OPEN alarm • Flare header FL-100 and flare tip	2	2	**4 L**	—	—
N11.3	Temperature	**LESS**	Cold flare line during blowdown	Expansion cooling of the gas through the BDVs and in the emptying vessels	Flare line and BDV outlets cool to about −21 °C in the model: brittle fracture of carbon steel, hydrates in the orifices.	Flare lines LTCS, MDMT −46 °C (KC-PID-100 note 3, TBC)	4	2	**8 M**	R32	Materials
N11.4	Pressure	**MORE**	Flare header overloaded	PV-102 flaring the full production after a trip while ESD-2 blows down	High back-pressure on PSV-001 and the BDVs, flare radiation on the deck, liquid carry-over from the knock-out drum.	• Flare header FL-100 and flare tip • FI-109 flare flow HI 5 t/h	4	2	**8 M**	R01	Process / Safety
N11.5	Timing	**LATE**	Blowdown started late	• ESD-2 is a manual decision • operator unsure or busy with the trip	A fire on the module heats a pressurised loop: escalation before the inventory is removed.	• Operator ESD-1 / ESD-2 on the SIS screen • Fire and gas detection and ESD (not on KC-PID-100; assumed)	4	2	**8 M**	R34	Operations / Safety
N11.6	Flow	**AS WELL AS**	Vent left open	HV-102 left open after venting the loop	Continuous flaring; on restart the loop cannot be held at pressure; isolation incomplete.	• FI-109 flare flow HI 5 t/h • HIC-102 output on the DCS	2	3	**6 M**	R07	Instrument
N11.7	Composition	**AS WELL AS**	Liquid to flare	Scrubber liquids carried through the BDVs or PV-102	Burning liquid rain from the flare tip on the platform or the sea.	Flare knock-out drum (outside the scope; assumed)	4	2	**8 M**	R33	Process
N11.8	Ignition	**NO**	Flare not lit	Pilot flame-out, ignition system failure	Unburnt gas cloud from the flare tip drifting over the platform: explosion if it finds an ignition source.	Flare pilots (assumed)	4	2	**8 M**	R33	Process

Node N12 · Procedures

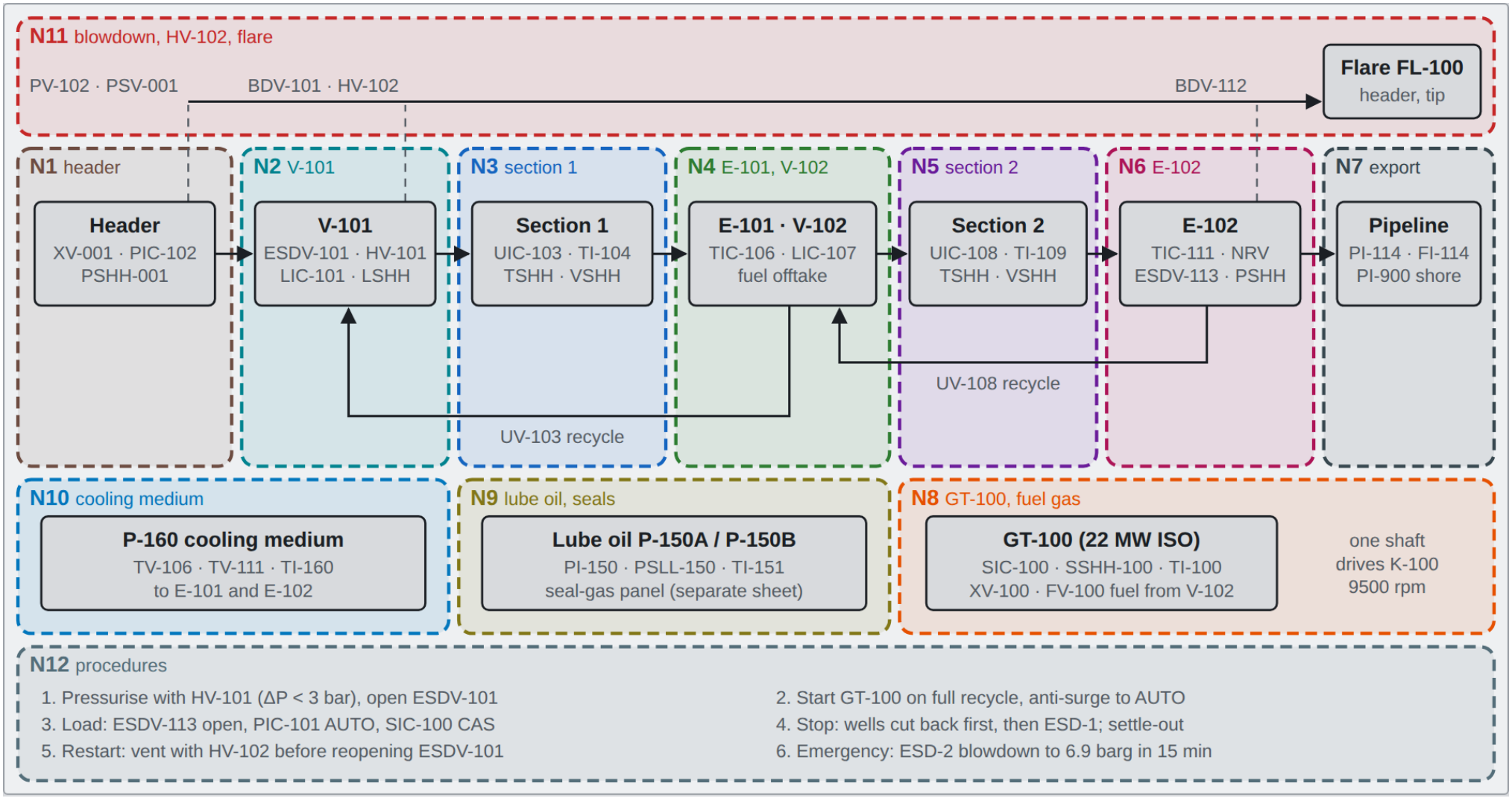
Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Node**	N12 · Procedures: pressurising, GT start, loading, settle-out, restart, blowdown
Design intent	Bring K-100 from a depressurised loop to full load and back safely: pressurise with HV-101, open ESDV-101, start GT-100 on full recycle, load with SIC-100 in cascade, stop with ESD-1 after the wells are cut back, vent the settled-out loop with HV-102 before a restart, and blow down with ESD-2 in an emergency.	**Design conditions**	Start-up limit: GT at minimum governor speed 70 %; reset needs the train below 5 %; ESDV-101 ΔP < 3 bar; blowdown to 6.9 barg in 6.1 min (model)
Equipment	Whole unit; Exercises tab procedures (cold start-up, normal shutdown, ESD with blowdown, GT trip restart)	**Instruments**	SIS screen (reset, ESD-1, ESD-2, ESDV commands, GT start), HIC-101, HIC-102
Drawing	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Document**	KC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N12.1	Sequence	**BEFORE**	ESDV-101 opened before pressurising	Operator tries to open ESDV-101 on a depressurised loop	Gas rushes into the empty loop: high velocity, strong cooling, mechanical shock to the scrubber internals.	• ESDV-101 open permissive: ΔP across the valve < 3 bar (pressurise with HV-101 first) • Written procedures (Exercises tab: cold start-up, normal shutdown, ESD with blowdown, GT trip restart)	3	1	**3 L**	—	—
N12.2	Sequence	**AFTER**	GT started before the recycles are open	Recycle valves not driven to 100 % before the start	Surge during acceleration (little flow, rising speed).	• GT start permissives (SIS reset, PI-150 ≥ 2.0 barg, P-160 running, ESDV-101 open with suction ≥ 20 barg, UV-103/108 ≥ 90 %, scrubber levels < 60 %, BDVs closed) and the 30 s crank-purge before ignition • SURGE alarms UIC-103 / UIC-108 (flow reversal)	3	2	**6 M**	—	—
N12.3	Timing	**EARLY**	Loaded before the anti-surge loops are in AUTO	SIC-100 put in cascade or speed raised with UIC-103/108 still in MAN	Surge on loading: thrust reversal, seal damage.	• SURGE alarms UIC-103 / UIC-108 (flow reversal) • VSHH-105 at 90 µm for 3 s → I-1 • VSHH-110 at 90 µm for 3 s → I-1	3	3	**9 M**	R10, R36	Instrument / Operations · Operations
N12.4	Timing	**LATE**	ESDV-113 opened late during loading	Speed raised against the closed export valve	Discharge packs toward PSHH-112: trip on start-up.	• PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1	2	3	**6 M**	—	—
N12.5	Restart	**OTHER THAN**	Restart without venting the settled-out loop	HV-101 or ESDV-101 opened at settle-out pressure instead of venting with HV-102 first	Gas at ≈ 69 barg back into the 30-barg header: PSV-001 lifts, PSHH-001 shuts the wells.	• ESDV-101 open permissive: ΔP across the valve < 3 bar (pressurise with HV-101 first) • PSHH-001 at 34.5 barg for 3 s → XV-001 shuts the wells • PSV-001 set 36 barg on the production header → flare	3	3	**9 M**	R36	Operations

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N12.6	Restart	**BEFORE**	Restart before the trip cause is cleared	Reset and start with the standby lube-oil pump not ready, a slug still in V-101, or the GT fault not cleared	Repeat trip or a damaged machine.	• SIS reset refused while a trip condition stands or the train still turns (> 5 %) • GT start permissives (SIS reset, PI-150 ≥ 2.0 barg, P-160 running, ESDV-101 open with suction ≥ 20 barg, UV-103/108 ≥ 90 %, scrubber levels < 60 %, BDVs closed) and the 30 s crank-purge before ignition	3	2	**6 M**	—	—
N12.7	Shutdown	**AFTER**	Compressor stopped before the wells are cut back	ESD-1 used before production is reduced	The full production (222 t/h) goes to flare (lost sales, CO_2, radiation).	• Written procedures (Exercises tab: cold start-up, normal shutdown, ESD with blowdown, GT trip restart) • Trip feed-forward: on I-1 PIC-102 output goes to ≥ 80 % (production flared at once)	2	3	**6 M**	R35	Operations / Process
N12.8	Timing	**LATE**	Slow recovery after a trip	Venting, reset and restart take hours	Production flared at about 222 t/h for the whole outage unless the wells are cut back.	Written procedures (Exercises tab: cold start-up, normal shutdown, ESD with blowdown, GT trip restart)	2	4	**8 M**	R35	Operations / Process
N12.9	Blowdown	**OTHER THAN**	ESD-2 used where ESD-1 was meant	• Wrong push-button • unclear criteria for blowing down	Unnecessary blowdown: the whole loop inventory flared, cold flare lines, a longer restart (re-pressurising from about 0 barg).	• Operator ESD-1 / ESD-2 on the SIS screen • Written procedures (Exercises tab: cold start-up, normal shutdown, ESD with blowdown, GT trip restart)	2	3	**6 M**	R34	Operations / Safety
N12.10	Maintenance	**PART OF**	Isolation by the ESDVs alone	Work on the compressor with only ESDV-101/ESDV-113 shut and the BDVs open	Gas passing a closed ESDV reaches the work site: fire or asphyxiation of the maintenance crew.	Permit to work (assumed)	4	2	**8 M**	R08	Operations / Mechanical

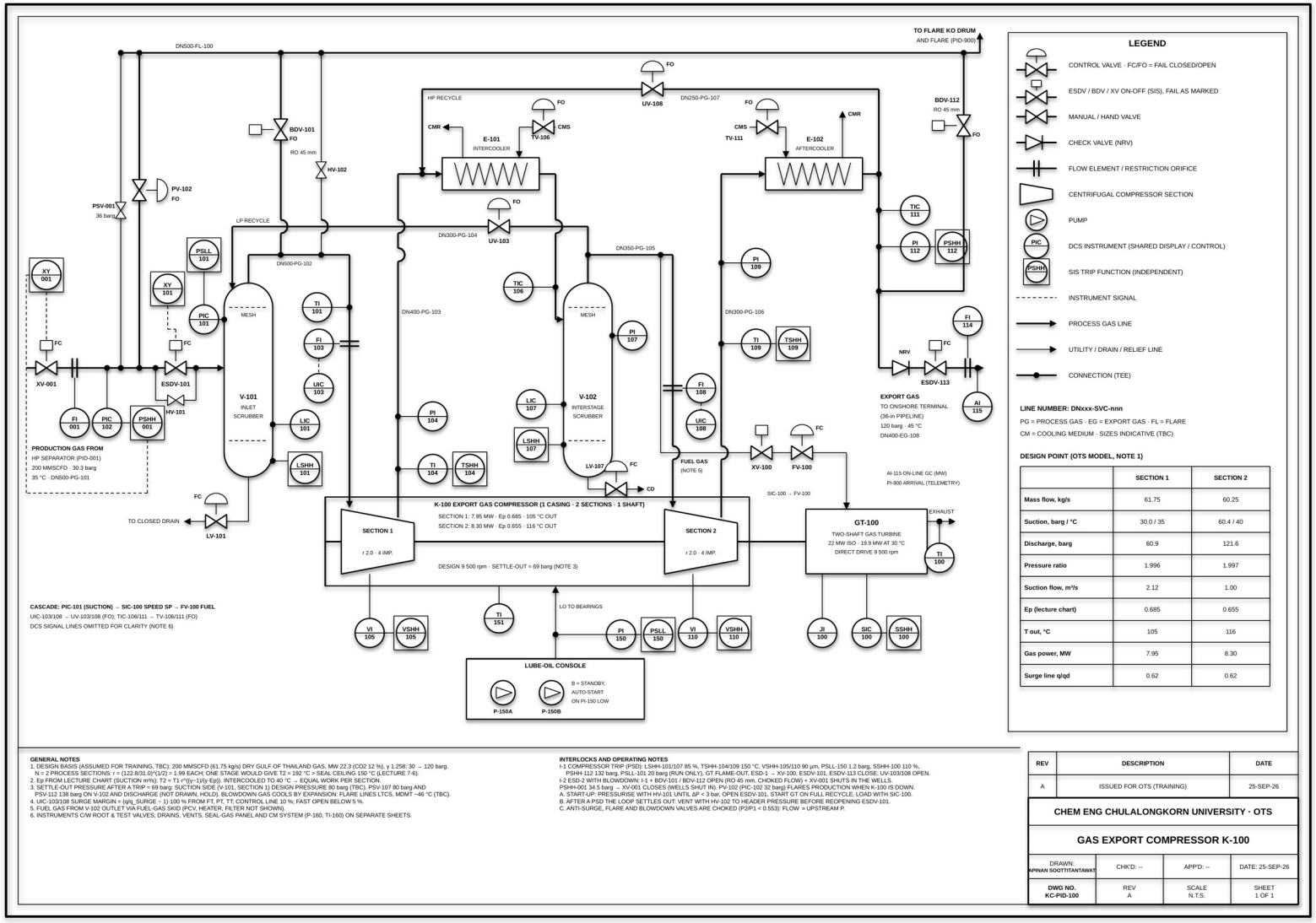
Appendix 2 · HAZOP documents (drawings)

A2-1 · HAZOP node map



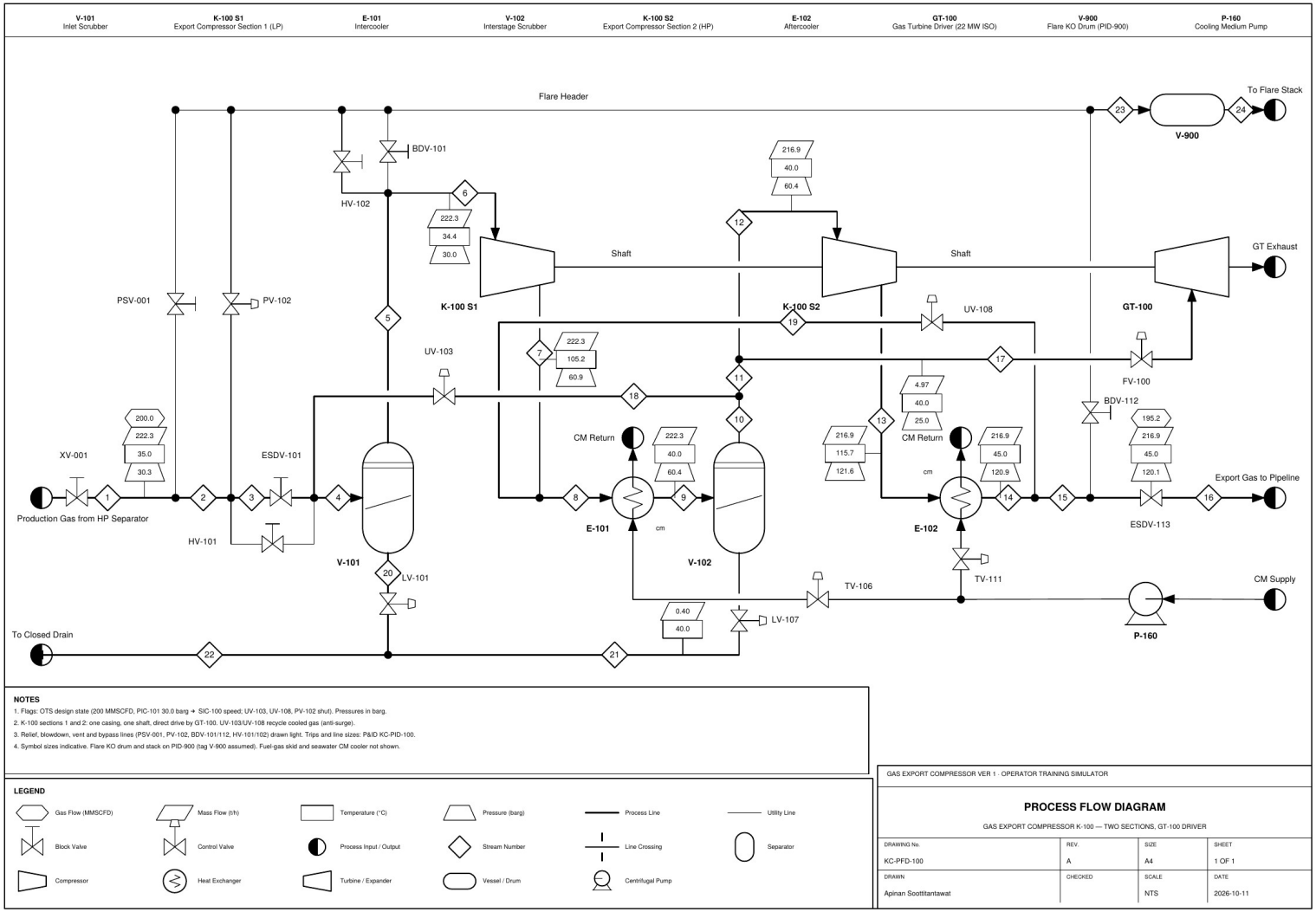
A2-1. HAZOP node map of the export compression train (nodes N1–N11; N12 = procedures) — the HAZOP tab, section 3.

A2-2 · KC-PID-100 rev A (P&ID)



A2-2. P&ID KC-PID-100 rev A: production header, V-101, K-100 sections 1 and 2, E-101, V-102, E-102, ESDV-113, GT-100, lube oil, cooling medium, blowdown and flare (P&ID tab). PSV-107 and PSV-112 are shown on hold (note 3).

A2-3 · KC-PFD-100 rev A (process flow diagram)



A2-3. Process flow diagram KC-PFD-100 rev A with the stream numbers of the Process tab (PFD tab).

Appendix 3 · HAZOP action close-out sheets

One sheet per recommendation (38 sheets). The action party writes the response, ticks the status and signs; the HAZOP leader accepts the close-out.

HAZOP ACTION CLOSE-OUT SHEET · R01

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R01**
Node	N1 / N11 · Production header / Blowdown BDV-101 / BDV-112	**Action party**	Process / Safety
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.4	Pressure **MORE**: Header overpressure when K-100 stops	K-100 trips (any I-1 cause) with PV-102 failing to open: PIC-102 left in MAN at a low output, PV-102 stuck, flare line blocked	Full production into a closed 150 m³ header: the pressure climbs toward the wells' shut-in pressure (59 barg in the model) — far above the header rating; flange or vessel failure, large gas release on the module, fire or explosion. With PSHH-001 also failed the model header peaks at 36.5 barg with PSV-001 relieving.	• PIC-102 HI 32.5 barg • Trip feed-forward: on I-1 PIC-102 output goes to ≥ 80 % (production flared at once) • PSHH-001 at 34.5 barg for 3 s → XV-001 shuts the wells • PSV-001 set 36 barg on the production header → flare	S5 L2 **10 High**
N11.4	Pressure **MORE**: Flare header overloaded	PV-102 flaring the full production after a trip while ESD-2 blows down	High back-pressure on PSV-001 and the BDVs, flare radiation on the deck, liquid carry-over from the knock-out drum.	• Flare header FL-100 and flare tip • FI-109 flare flow HI 5 t/h	S4 L2 **8 Medium**

HAZOP recommendation
Relief and flare study: confirm PSV-001 for the blocked-outlet case with the full production (K-100 tripped and PV-102 shut), and the flare header and tip for PV-102 at full production plus simultaneous blowdown (back-pressure on PSV-001 and the BDVs, radiation on the deck).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R02

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R02**
Node	N1 · Production header	**Action party**	Mechanical / Operations
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.5	Pressure **LESS**: Header pressure falls: gas to flare	• PV-102 fails open (loss of air, fail-open design) • PV-102 seat passing (fault library) • PIC-102 set point entered too low	Production gas burnt in the flare instead of exported (lost sales and CO_2); suction pressure falls, K-100 slows and can trip on PSLL-101.	• FI-109 flare flow HI 5 t/h • PIC-101 LO 28.5 / HI 31.5 barg • PSLL-101 at 20 barg for 5 s → I-1 (armed in RUN)	S2 L3 **6 Medium**

HAZOP recommendation
Specify the seat-leak class of PV-102 and test it at each shutdown; report flared gas as a loss figure (valued at the training gas price).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R03

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R03**
Node	N1 · Production header	**Action party**	Instrument
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.4	Pressure **MORE**: Header overpressure when K-100 stops	K-100 trips (any I-1 cause) with PV-102 failing to open: PIC-102 left in MAN at a low output, PV-102 stuck, flare line blocked	Full production into a closed 150 m³ header: the pressure climbs toward the wells' shut-in pressure (59 barg in the model) — far above the header rating; flange or vessel failure, large gas release on the module, fire or explosion. With PSHH-001 also failed the model header peaks at 36.5 barg with PSV-001 relieving.	• PIC-102 HI 32.5 barg • Trip feed-forward: on I-1 PIC-102 output goes to ≥ 80 % (production flared at once) • PSHH-001 at 34.5 barg for 3 s → XV-001 shuts the wells • PSV-001 set 36 barg on the production header → flare	S5 L2 **10 High**

HAZOP recommendation
Give PSHH-001 its own pressure transmitter: today PT-001 serves the PI-001 indication, PIC-102 and PSHH-001.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R04

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R04**
Node	N2 · Inlet scrubber V-101	**Action party**	Process / Operations
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.1	Level **MORE**: Liquid carried into section 1	• Liquid slug from production (fault library) • LV-101 stuck shut or the closed drain blocked • LIC-101 left in MAN	V-101 fills (with no trip the model reaches 100 % 82 s after the largest slug starts) and liquid enters section 1: impeller erosion or breakage, rotor imbalance, seal failure, gas release.	• LIC-101 → LV-101 to closed drain (fail closed) • LIC-101 LO 20 / HI 65 % • LSHH-101 at 85 % for 3 s → I-1	S4 L3 **12 High**

HAZOP recommendation
Slug handling: check LV-101 and the closed drain against the largest credible slug and agree a slug warning from production (well tests, pigging, restarts).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R05

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R05**
Node	N2 / N4 · Inlet scrubber V-101 / Intercooler E-101	**Action party**	Process
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.2	Level **LESS**: Gas blow-by to the closed drain	• LV-101 fails open or LIC-101 output high • level transmitter reads high	V-101 empties and gas at 30 barg flows into the low-pressure closed drain: drain drum overpressure, gas release at the drain vent.	• LIC-101 → LV-101 to closed drain (fail closed) • LIC-101 LO 20 / HI 65 % • Closed-drain system (separate sheet; assumed)	S4 L2 **8 Medium**
N4.4	Level **LESS**: Gas blow-by from V-102 to the closed drain	• LV-107 fails open • LIC-107 output high	Gas at 60 barg into the low-pressure closed drain: drain drum overpressure, gas release.	• LIC-107 → LV-107 to closed drain (fail closed) • LIC-107 LO 20 / HI 65 % • Closed-drain system (separate sheet; assumed)	S4 L2 **8 Medium**

HAZOP recommendation
Gas blow-by protection of the closed drain: low-low level on V-101 and V-102 that closes LV-101 / LV-107, and relief of the drain drum sized for gas blow-by through a wide-open LV.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R06

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R06**
Node	N2 · Inlet scrubber V-101	**Action party**	Instrument
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.3	Pressure **LESS**: Suction pressure low, unseen	• PIC-101 transmitter freezes or drifts (fault library) • ESDV-101 partly shut	Speed control acts on a false pressure: the real suction can fall without alarm, the ratio and the discharge temperatures rise, the margins shrink. PSLL-101 reads the same measurement, so the trip is blind too.	• PIC-101 LO 28.5 / HI 31.5 barg (same transmitter) • PSLL-101 at 20 barg for 5 s → I-1 (armed in RUN) (same transmitter) • TSHH-104 at 150 °C for 3 s → I-1	S3 L3 **9 Medium**

HAZOP recommendation
Put PSLL-101 on its own transmitter: the trip reads the PIC-101 measurement, so a frozen or drifting PIC-101 transmitter blinds the control loop, the alarm and the trip together.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R07

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R07**
Node	N2 / N11 · Inlet scrubber V-101 / Blowdown BDV-101 / BDV-112	**Action party**	Instrument
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.7	Flow **AS WELL AS**: Flow through the bypass after isolation	HV-101 left open after start-up	ESDV-101 shuts on a trip but gas keeps entering the loop through HV-101: isolation and blowdown are defeated (the BDV fights the inflow).	• HIC-101 output on the DCS • Written procedures (Exercises tab: cold start-up, normal shutdown, ESD with blowdown, GT trip restart)	S3 L3 **9 Medium**
N11.6	Flow **AS WELL AS**: Vent left open	HV-102 left open after venting the loop	Continuous flaring; on restart the loop cannot be held at pressure; isolation incomplete.	• FI-109 flare flow HI 5 t/h • HIC-102 output on the DCS	S2 L3 **6 Medium**

HAZOP recommendation
Position switches on HV-101 and HV-102 with an alarm when either is open while the unit is in RUN.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R08

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R08**
Node	N12 · Procedures	**Action party**	Operations / Mechanical
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N12.10	Maintenance **PART OF**: Isolation by the ESDVs alone	Work on the compressor with only ESDV-101/ESDV-113 shut and the BDVs open	Gas passing a closed ESDV reaches the work site: fire or asphyxiation of the maintenance crew.	Permit to work (assumed)	S4 L2 **8 Medium**

HAZOP recommendation
Positive isolation for work on the compressor: double block-and-bleed or spades at the ESDVs; ESDV-101 and ESDV-113 leak tests at each proof test.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R09

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R09**
Node	N3 / N5 · Section 1 (LP) and UV-103 recycle / Section 2 (HP) and UV-108 recycle	**Action party**	Instrument
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.2	Measurement **OTHER THAN**: Anti-surge flow reads wrong	• FT-103 reads high (fault library) • impulse line plugged or leaking	UIC-103 believes the margin is larger than it is: no recycle until the machine is already in surge on the next turndown.	• SURGE alarms UIC-103 / UIC-108 (flow reversal) • VSHH-105 at 90 µm for 3 s → I-1	S4 L3 **12 High**
N5.2	Flow **LESS**: Section 2 driven into surge	• Production cut • arrival pressure rising (fault library) • UIC-108 in MAN or slow	Surge in the denser HP section: violent flow reversals, thrust reversal, seal and bearing failure, gas release at 60–120 barg.	• UIC-108 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-108 cooled recycle, fail open • UIC-108 margin LO 7 % and SURGE alarm • VSHH-110 at 90 µm for 3 s → I-1 • TSHH-109 at 150 °C for 3 s → I-1	S4 L3 **12 High**

HAZOP recommendation
Validate the anti-surge flow measurements: second flow element or a mass-balance deviation alarm, because UIC-103 and UIC-108 each act on one transmitter.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R10

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R10**
Node	N3 / N5 / N12 · Section 1 (LP) and UV-103 recycle / Section 2 (HP) and UV-108 recycle / Procedures	**Action party**	Instrument / Operations
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.3	Control **OTHER THAN**: Anti-surge controller in MAN	UIC-103 left in MAN after start-up, a valve stroke or tuning	No surge protection on turndown. With the wells cutting back and both loops in MAN the model surges 42 times in 5 min and TI-104 reaches 171 °C (no-trip run).	• SURGE alarms UIC-103 / UIC-108 (flow reversal) • UIC-103 margin LO 7 % and SURGE alarm • TSHH-104 at 150 °C for 3 s → I-1 • VSHH-105 at 90 µm for 3 s → I-1	S4 L3 **12 High**
N5.2	Flow **LESS**: Section 2 driven into surge	• Production cut • arrival pressure rising (fault library) • UIC-108 in MAN or slow	Surge in the denser HP section: violent flow reversals, thrust reversal, seal and bearing failure, gas release at 60–120 barg.	• UIC-108 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-108 cooled recycle, fail open • UIC-108 margin LO 7 % and SURGE alarm • VSHH-110 at 90 µm for 3 s → I-1 • TSHH-109 at 150 °C for 3 s → I-1	S4 L3 **12 High**
N12.3	Timing **EARLY**: Loaded before the anti-surge loops are in AUTO	SIC-100 put in cascade or speed raised with UIC-103/108 still in MAN	Surge on loading: thrust reversal, seal damage.	• SURGE alarms UIC-103 / UIC-108 (flow reversal) • VSHH-105 at 90 µm for 3 s → I-1 • VSHH-110 at 90 µm for 3 s → I-1	S3 L3 **9 Medium**

HAZOP recommendation
Alarm when UIC-103 or UIC-108 is not in AUTO with the unit in RUN; manual mode by supervisor access only.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R11

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R11**
Node	N3 / N5 · Section 1 (LP) and UV-103 recycle / Section 2 (HP) and UV-108 recycle	**Action party**	Instrument
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.1	Flow **LESS**: Section 1 driven into surge	• Fast production cut • UV-103 sticks (fault library) • anti-surge loop too slow or badly tuned	Surge: forward flow collapses and reverses every few seconds, thrust reverses, temperature builds up in the loop; seals and thrust bearing fail, rotor rub, gas release at the seals.	• UIC-103 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-103 cooled recycle, fail open • UIC-103 margin LO 7 % and SURGE alarm • VSHH-105 at 90 µm for 3 s → I-1 • TSHH-104 at 150 °C for 3 s → I-1	S4 L3 **12 High**
N3.4	Flow **NO**: No recycle when the trip demands it	UV-103 stuck at its running position (fault library) when I-1 calls for 100 %	The coast-down follows a falling speed with no recycle: surge cycles during the rundown, thrust-bearing and seal damage.	• On I-1 UV-103 and UV-108 go to 100 % (trip opening) (fails with the stuck valve) • SURGE alarms UIC-103 / UIC-108 (flow reversal)	S3 L2 **6 Medium**
N5.1	Flow **MORE**: Large HP recycle	• UV-108 fails open (fault library: air or solenoid failure) • UIC-108 output high	Most of the section 2 flow recirculates: export falls sharply, GT-100 goes to its power limit, suction rises and PV-102 flares production; the machine runs at its top speed near stonewall.	• FI-114 export LO 150 MMSCFD • GT power-limit alarm (FV-100 ≥ 98 %) • PIC-102 (SP 32 barg) → PV-102 to flare, fail open • FI-109 flare flow HI 5 t/h • SSHH-100 at 110 % for 0.5 s → I-1	S2 L3 **6 Medium**
N5.6	Timing **LATE**: Recycle opens too slowly on a trip	Slow actuator, undersized solenoid or exhaust on UV-108	Section 2 surges in the first seconds of the coast-down (the speed falls faster than the recycle opens).	• On I-1 UV-103 and UV-108 go to 100 % (trip opening) • SURGE alarms UIC-103 / UIC-108 (flow reversal)	S3 L2 **6 Medium**

HAZOP recommendation
Partial-stroke and stroke-time tests of UV-103 and UV-108; valve-position deviation alarm (ZI against controller output).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R12

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R12**
Node	N1 / N2 / N3 / N5 · Production header / Inlet scrubber V-101 / Section 1 (LP) and UV-103 recycle / Section 2 (HP) and UV-108 recycle	**Action party**	Instrument / Rotating
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.2	Flow **LESS**: Production falls fast	• Wells cut back (fault library: production falls by up to 55 %) • well work • choke change upstream	K-100 turns down; both operating points move toward surge. If the anti-surge loops do not keep up the sections surge: flow reversal, thrust reversal, seal and bearing damage, possible gas release at the seals.	• UIC-103 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-103 cooled recycle, fail open • UIC-108 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-108 cooled recycle, fail open • UIC-103 margin LO 7 % and SURGE alarm • SURGE alarms UIC-103 / UIC-108 (flow reversal) • VSHH-105 at 90 µm for 3 s → I-1	S4 L2 **8 Medium**
N2.6	Flow **NO**: Suction isolated while running	ESDV-101 closes spuriously (solenoid, ESD signal fault)	Suction pressure collapses within seconds; the sections are driven toward surge during the trip and the rundown.	• PSLL-101 at 20 barg for 5 s → I-1 (armed in RUN) • UIC-103 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-103 cooled recycle, fail open • On I-1 UV-103 and UV-108 go to 100 % (trip opening)	S3 L2 **6 Medium**
N3.1	Flow **LESS**: Section 1 driven into surge	• Fast production cut • UV-103 sticks (fault library) • anti-surge loop too slow or badly tuned	Surge: forward flow collapses and reverses every few seconds, thrust reverses, temperature builds up in the loop; seals and thrust bearing fail, rotor rub, gas release at the seals.	• UIC-103 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-103 cooled recycle, fail open • UIC-103 margin LO 7 % and SURGE alarm • VSHH-105 at 90 µm for 3 s → I-1 • TSHH-104 at 150 °C for 3 s → I-1	S4 L3 **12 High**
N3.2	Measurement **OTHER THAN**: Anti-surge flow reads wrong	• FT-103 reads high (fault library) • impulse line plugged or leaking	UIC-103 believes the margin is larger than it is: no recycle until the machine is already in surge on the next turndown.	• SURGE alarms UIC-103 / UIC-108 (flow reversal) • VSHH-105 at 90 µm for 3 s → I-1	S4 L3 **12 High**
N5.2	Flow **LESS**: Section 2 driven into surge	• Production cut • arrival pressure rising (fault library) • UIC-108 in MAN or slow	Surge in the denser HP section: violent flow reversals, thrust reversal, seal and bearing failure, gas release at 60–120 barg.	• UIC-108 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-108 cooled recycle, fail open • UIC-108 margin LO 7 % and SURGE alarm • VSHH-110 at 90 µm for 3 s → I-1 • TSHH-109 at 150 °C for 3 s → I-1	S4 L3 **12 High**

HAZOP recommendation
Independent surge detection: surge counter that trips I-1 on repeated surge cycles (for example 3 in 30 s), with the cycles logged for the rotating-equipment engineer.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R13

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R13**
Node	N2 / N4 / N6 / N11 · Inlet scrubber V-101 / Intercooler E-101 / Aftercooler E-102 / Blowdown BDV-101 / BDV-112	**Action party**	Process
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.4	Pressure **MORE**: Suction pressure high	• K-100 at its power or speed limit • PIC-101 in MAN • settle-out after a trip	Production is flared through PV-102; after a trip the whole loop settles out at ≈ 69 barg in V-101 (model), more than twice the header pressure.	• PIC-101 LO 28.5 / HI 31.5 barg • PIC-102 (SP 32 barg) → PV-102 to flare, fail open • Suction side designed for 80 barg (KC-PID-100 note 3, TBC), above the settle-out pressure	S2 L3 **6 Medium**
N4.5	Pressure **MORE**: Interstage at settle-out after a trip	Any I-1 trip: the loop equalises through the open recycle valves	E-101 and V-102 sit at the settle-out pressure (≈ 69 barg in the model) instead of 60 barg; a fire on the module would raise it further with no relief installed.	• Suction side designed for 80 barg (KC-PID-100 note 3, TBC), above the settle-out pressure • PSV-107 80 barg on V-102 (KC-PID-100 note 3: on hold, not drawn — not credited until installed) • ESD-2: BDV-101 and BDV-112 open to flare (fail open, RO 45 mm, choked), XV-001 shuts the wells	S4 L2 **8 Medium**
N6.1	Flow **NO**: Discharge blocked at full load	• ESDV-113 closes spuriously (fault library: solenoid fault) • downstream block valve shut • NRV stuck shut	The 30 m³ discharge header packs at full GT power: with no trip the model passes 138 barg 11 s after the closure and reaches 178 barg in 60 s — beyond the discharge design; rupture and a high-pressure gas release, fire or explosion.	• ESDV-113 SHUT alarm with the compressor running • PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1 • PSV-112 138 barg on the discharge (KC-PID-100 note 3: on hold, not drawn — not credited until installed)	S5 L2 **10 High**
N11.1	Flow **NO**: Blowdown fails	BDV stuck shut, solenoid fails energised, restriction orifice blocked by hydrate	The loop stays at the settle-out pressure (≈ 69 barg) during a fire: vessel or pipe ruptures when the steel weakens; escalation.	• ESD-2: BDV-101 and BDV-112 open to flare (fail open, RO 45 mm, choked), XV-001 shuts the wells • BDV OPEN alarm • PSV-107 80 barg on V-102 (KC-PID-100 note 3: on hold, not drawn — not credited until installed)	S5 L2 **10 High**

HAZOP recommendation
Confirm and install PSV-107 (80 barg, V-102) and PSV-112 (138 barg, discharge), on hold on KC-PID-100: size for settle-out with fire, and for a blocked outlet at full GT power.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R14

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R14**
Node	N6 · Aftercooler E-102	**Action party**	Instrument / Process
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.1	Flow **NO**: Discharge blocked at full load	• ESDV-113 closes spuriously (fault library: solenoid fault) • downstream block valve shut • NRV stuck shut	The 30 m³ discharge header packs at full GT power: with no trip the model passes 138 barg 11 s after the closure and reaches 178 barg in 60 s — beyond the discharge design; rupture and a high-pressure gas release, fire or explosion.	• ESDV-113 SHUT alarm with the compressor running • PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1 • PSV-112 138 barg on the discharge (KC-PID-100 note 3: on hold, not drawn — not credited until installed)	S5 L2 **10 High**

HAZOP recommendation

Review PSHH-112: set point and delay against PSV-112 (the dynamic check overshoots past the trip setting before the machine unloads); consider a shorter delay or a lower setting.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R15

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R15**
Node	N6 · Aftercooler E-102	**Action party**	Instrument / Mechanical
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.1	Flow **NO**: Discharge blocked at full load	• ESDV-113 closes spuriously (fault library: solenoid fault) • downstream block valve shut • NRV stuck shut	The 30 m³ discharge header packs at full GT power: with no trip the model passes 138 barg 11 s after the closure and reaches 178 barg in 60 s — beyond the discharge design; rupture and a high-pressure gas release, fire or explosion.	• ESDV-113 SHUT alarm with the compressor running • PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1 • PSV-112 138 barg on the discharge (KC-PID-100 note 3: on hold, not drawn — not credited until installed)	S5 L2 **10 High**
N6.5	Flow **REVERSE**: Pipeline gas back into the compressor	After a trip the NRV passes and ESDV-113 fails to close	Line-pack (450 m³ at about 120 barg) flows back through E-102 and section 2: reverse rotation, the loop rises toward pipeline pressure above the suction-side design, fire if the seals fail.	• Check valve (NRV) upstream of ESDV-113 • ESDV-113 shuts on I-1 and ESD • ESD-2: BDV-101 and BDV-112 open to flare (fail open, RO 45 mm, choked), XV-001 shuts the wells	S4 L2 **8 Medium**

HAZOP recommendation
ESDV-113: reduce spurious closure (redundant solenoids), proof-test closure time and seat leakage; inspect the NRV at each shutdown.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R16

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R16**
Node	N7 · Export riser and pipeline to the onshore terminal	**Action party**	Pipeline
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N7.1	Pressure **MORE**: Line-pack rises	• Onshore terminal restricts its intake (fault library: arrival pressure rises by up to 30 bar) • terminal pressure control fault	Pipeline and discharge pressure climb until PSHH-112 trips K-100; with no trip the model reaches 139 barg in 30 min — above the PSV-112 setting and possibly the pipeline MAOP: rupture of the line or riser.	• PI-900 arrival pressure telemetry from the onshore terminal (indication only, no alarm) • PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1	S4 L3 **12 High**

HAZOP recommendation
Confirm the export pipeline MAOP against the highest discharge pressure (PSV-112 set point); if lower, add pipeline overpressure protection (HIPPS).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R17

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R17**
Node	N6 / N7 · Aftercooler E-102 / Export riser and pipeline to the onshore terminal	**Action party**	Operations / Instrument
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.2	Pressure **MORE**: Discharge pressure rising slowly	• Onshore arrival pressure rising (fault library) • production above design with the pipeline packing	K-100 works harder until PSHH-112 trips it: export lost and production flared until the wells are cut back.	• PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1 • PI-900 arrival pressure telemetry from the onshore terminal (indication only, no alarm)	S3 L3 **9 Medium**
N7.1	Pressure **MORE**: Line-pack rises	• Onshore terminal restricts its intake (fault library: arrival pressure rises by up to 30 bar) • terminal pressure control fault	Pipeline and discharge pressure climb until PSHH-112 trips K-100; with no trip the model reaches 139 barg in 30 min — above the PSV-112 setting and possibly the pipeline MAOP: rupture of the line or riser.	• PI-900 arrival pressure telemetry from the onshore terminal (indication only, no alarm) • PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1	S4 L3 **12 High**
N7.2	Flow **NO**: Onshore intake shut	• Terminal inlet ESD valve closes • terminal blackout	The pipeline packs at the full export rate; PSHH-112 trips K-100 and production is flared until the wells are cut back.	• PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1 • Trip feed-forward: on I-1 PIC-102 output goes to ≥ 80 % (production flared at once)	S3 L2 **6 Medium**

HAZOP recommendation
Agreement with the onshore terminal on intake restrictions with advance notice; bring PI-900 into the DCS with a high alarm.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R18

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R18**
Node	N7 · Export riser and pipeline to the onshore terminal	**Action party**	Pipeline / Safety
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N7.3	Integrity **LESS**: Pipeline or riser rupture	Anchor drag or dropped object, corrosion (wet CO_2), riser damage by a vessel	Massive subsea or riser gas release: a gas plume at the sea surface or a riser fire under the platform; the platform has no isolation between the riser and the line-pack except ESDV-113.	• ESDV-113 shuts on I-1 and ESD • PI-114 / FI-114 indications	S5 L2 **10 High**

HAZOP recommendation
Riser ESD valve or subsea isolation valve near the platform and pipeline leak detection (pressure/flow balance with the terminal).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R19

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R19**
Node	N1 / N6 / N7 · Production header / Aftercooler E-102 / Export riser and pipeline to the onshore terminal	**Action party**	Process / Operations
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.7	Composition **OTHER THAN**: Richer (heavier) gas	New well with heavier gas (fault library: MW 22.3 → 24.7)	Lower speed of sound and more condensate in the coolers: higher V-102 level, liquids toward the fuel gas; the export gas can leave its hydrocarbon dew-point specification; volumetric meters read low.	• AI-115 molar mass HI 23.5 kg/kmol • LIC-107 → LV-107 to closed drain (fail closed) • LIC-107 LO 20 / HI 65 % • LSHH-107 at 85 % for 3 s → I-1	S2 L3 **6 Medium**
N6.4	Temperature **LESS**: Export gas too cold	• TV-111 fails open with cold cooling medium • TIC-111 set point too low	Export gas below its hydrocarbon dew point: liquids in the pipeline, slugs at the onshore terminal.	• TIC-111 (SP 45 °C) → TV-111 fail open • TIC-111 LO 30 / HI 55 °C	S2 L2 **4 Low**
N7.5	Composition **OTHER THAN**: Export gas off specification	• Richer gas from a new well (fault library) • export too cold (node N6)	Hydrocarbon dew point exceeded: liquids in the pipeline, sales gas rejected at the terminal.	• AI-115 molar mass HI 23.5 kg/kmol • TIC-111 (SP 45 °C) → TV-111 fail open • TIC-111 LO 30 / HI 55 °C	S2 L3 **6 Medium**

HAZOP recommendation
Export gas quality: hydrocarbon dew-point limit on TIC-111 and the export gas, and a response plan for an AI-115 high alarm (new wells).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R20

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R20**
Node	N1 / N7 · Production header / Export riser and pipeline to the onshore terminal	**Action party**	Materials / Process
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.8	Composition **AS** **WELL AS**: Free water with CO_2-rich gas	• Dehydration upset upstream (outside the scope) • water carried from the HP separator	Wet CO_2 (12 mol %) corrodes carbon-steel header, coolers and pipeline; hydrates where the gas expands (HV-101, BDVs, PV-102).	• Upstream dehydration (outside the scope; assumed) • LIC-101 → LV-101 to closed drain (fail closed) • LIC-101 LO 20 / HI 65 %	S3 L2 **6 Medium**
N7.6	Composition **AS** **WELL AS**: Water in the export gas	Dehydration upset upstream (outside the scope)	Internal corrosion of the pipeline by wet CO_2, hydrates during shut-ins.	Upstream dehydration (outside the scope; assumed)	S4 L2 **8 Medium**

HAZOP recommendation
Corrosion management for wet CO_2 (12 mol %): water dew-point monitoring at the inlet, inhibitor, corrosion coupons in the header and the pipeline.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R21

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R21**
Node	N4 / N8 · Intercooler E-101 / GT-100 driver and fuel gas	**Action party**	Process
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.3	Level **MORE**: Liquid into section 2 and the fuel gas	• LV-107 sticks shut (fault library) • richer gas (fault library) • overcooled interstage gas	Liquid carry-over into section 2 (impeller damage) and into the GT fuel gas (flame instability, hot streaks, flame-out).	• LIC-107 → LV-107 to closed drain (fail closed) • LIC-107 LO 20 / HI 65 % • LSHH-107 at 85 % for 3 s → I-1	S4 L2 **8 Medium**
N4.6	Pressure **LESS**: Fuel gas pressure low	• Interstage pressure low during start-up, turndown or after a trip • fuel-gas skid restricted (fault library: fuel gas pressure low)	GT-100 loses available power or cannot accelerate; speed falls, suction rises and PV-102 flares; a flame-out trips the unit.	• GT power-limit alarm (FV-100 ≥ 98 %) • PIC-102 (SP 32 barg) → PV-102 to flare, fail open • GT-100 flame-out trips I-1 • XV-100 shuts the fuel	S2 L3 **6 Medium**
N8.5	Composition **AS WELL AS**: Liquids in the fuel gas	• V-102 carry-over (node N4) • richer gas condensing in the fuel line	Flame instability, hot streaks, combustor damage, flame-out (driver lost).	• LIC-107 → LV-107 to closed drain (fail closed) • LIC-107 LO 20 / HI 65 % • LSHH-107 at 85 % for 3 s → I-1	S3 L2 **6 Medium**

HAZOP recommendation
Fuel-gas conditioning (knock-out, superheater, filter — note 5) with a high-level trip, and a fuel source for start-up and low interstage pressure.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R22

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R22**
Node	N1 / N8 · Production header / GT-100 driver and fuel gas	**Action party**	Operations / Process
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.3	Flow **MORE**: Production above the design rate	• New well on line (fault library: production rises by up to 15 %) • well test diverted to the header	GT-100 reaches its power limit; suction pressure rises and PV-102 flares the excess; the discharge and the pipeline pack toward PSHH-112.	• GT power-limit alarm (FV-100 ≥ 98 %) • PIC-101 LO 28.5 / HI 31.5 barg • PIC-102 (SP 32 barg) → PV-102 to flare, fail open • PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1	S2 L4 **8 Medium**
N8.2	Power **LESS**: GT available power falls	• Hot ambient (fault library) • fuel-gas pressure low (fault library) • GT compressor fouling	GT at its limit: speed sags, suction rises above set point and PV-102 flares; with extra production the shortfall grows.	• GT power-limit alarm (FV-100 ≥ 98 %) • SIC-100 LO 72 / HI 104 % • PIC-102 (SP 32 barg) → PV-102 to flare, fail open	S2 L4 **8 Medium**

HAZOP recommendation
Operating envelope for the operators: production that K-100 can take against ambient temperature and fuel-gas pressure (GT available power).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R23

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R23**
Node	N8 · GT-100 driver and fuel gas	**Action party**	Instrument / Rotating
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N8.3	Speed **MORE**: Overspeed	• Sudden loss of load (coupling failure, large recycle with the governor slow) • SIC-100 or FV-100 failure	Rotor and impeller burst at excessive speed: fragments through the casing, fatal injuries. Normal top speed is 105 %; with UV-108 failed open the model reaches 105.5 %.	• SIC-100 LO 72 / HI 104 % • SSHH-100 at 110 % for 0.5 s → I-1 • FV-100 fuel valve fail closed • XV-100 fuel ESD valve	S5 L1 **5 Medium**

HAZOP recommendation
Overspeed protection: confirm the speed probes and voting of SSHH-100 and proof-test it with the GT supplier.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R24

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R24**
Node	N3 / N5 / N9 · Section 1 (LP) and UV-103 recycle / Section 2 (HP) and UV-108 recycle / Lube oil and seal system	**Action party**	Rotating
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.7	Vibration **MORE**: Rotor vibration high	Deposits, liquid carry-over, surge, misalignment	Bearing and seal damage, rub; a long run at high vibration ends in a seal failure and a gas release.	• VI-105 HI 60 µm • VSHH-105 at 90 µm for 3 s → I-1	S3 L2 **6 Medium**
N5.4	Vibration **MORE**: HP-end vibration and bearing temperature rise	• Section 2 bearing wear (fault library) • imbalance • surge	Bearing failure, rub, seal failure and gas release; long repair.	• VI-110 HI 60 µm • VSHH-110 at 90 µm for 3 s → I-1 • TI-151 bearing HI 95 °C	S4 L2 **8 Medium**
N9.4	Temperature **MORE**: Bearing temperature high	• Bearing wear (fault library) • lube-oil cooler fouled or cooling medium hot • oil degraded	Bearing failure, rub, seal failure.	• TI-151 bearing HI 95 °C • VSHH-110 at 90 µm for 3 s → I-1	S3 L2 **6 Medium**

HAZOP recommendation
Bearing protection: high-high bearing temperature trip or condition monitoring on TI-151 (alarm only today).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R25

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R25**
Node	N9 · Lube oil and seal system	**Action party**	Rotating / Electrical
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N9.1	Pressure **LESS**: Main lube-oil pump stops	• P-150A trips (fault library) • motor or power fault	Without oil the bearings wipe within seconds at full speed: rotor damage, seal failure and gas release.	• PI-150 LO 1.8 barg • P-150B auto-starts at 1.8 barg (selector AUTO) • P-150B RUN alarm (main pump stopped) • PSLL-150 at 1.2 barg for 2 s → I-1	S4 L2 **8 Medium**
N9.2	Pressure **NO**: No lube oil	Main pump trips and the standby is not available on demand (fault library: standby fails to start)	PSLL-150 trips K-100, but the train needs about 7 min to stop with no oil: bearings run dry through the coast-down (model bearing temperature 126 °C after 2 min with no trip).	• PI-150 LO 1.8 barg • P-150B auto-starts at 1.8 barg (selector AUTO) • PSLL-150 at 1.2 barg for 2 s → I-1	S4 L3 **12 High**
N9.3	Power **NO**: Platform power failure	Main power lost: both AC lube-oil pumps stop together	As above: no lubrication during the coast-down.	PSLL-150 at 1.2 barg for 2 s → I-1	S4 L2 **8 Medium**

HAZOP recommendation
Lube oil: monitor the readiness of the standby pump (auto-start armed, pump healthy) with a DCS alarm; rundown tank (or DC emergency pump) that lubricates the bearings for the whole coast-down.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R26

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R26**
Node	N9 · Lube oil and seal system	**Action party**	Process / Rotating
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N9.5	Composition **AS** **WELL AS**: Gas or water in the lube oil	• Seal leakage into the oil • lube-oil cooler tube leak	Foaming, loss of oil film, bearing failure; flammable gas from the reservoir vent.	Oil sampling and reservoir vent (assumed)	S3 L2 **6 Medium**
N9.6	Flow **MORE**: Seal leakage high	Shaft-seal damage by surge thrust, liquid carry-over or discharge above 150 °C	Process gas to the seal vent or into the compressor house: fire or explosion.	• Seal-gas panel with its own alarms (separate sheet, not modelled; assumed) • TSHH-104 at 150 °C for 3 s → I-1 • TSHH-109 at 150 °C for 3 s → I-1	S4 L2 **8 Medium**
N9.7	Pressure **LESS**: Seal-gas supply lost	• No discharge pressure at start-up or after settle-out • seal-gas filter blocked	Unfiltered process gas reaches the seal faces: seal damage and a later leak.	Seal-gas panel with its own alarms (separate sheet, not modelled; assumed)	S3 L2 **6 Medium**

HAZOP recommendation
Extend the HAZOP to the seal-gas panel and the lube-oil console sheets: seal-gas supply at start-up and at settle-out, primary-vent high-flow trip, gas in the oil reservoir.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R27

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R27**
Node	N10 · Cooling medium	**Action party**	Process / Mechanical
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N10.1	Flow **NO**: Cooling medium lost	• P-160 trips (fault library) • header valve closed • power failure	No cooling in either cooler: section 2 suction and export temperatures climb within a minute; with no trip the model reaches TI-109 205 °C and export 202 °C — seal failure and pipeline coating damage.	• P-160 STOP alarm with the unit running • TIC-106 (SP 40 °C) → TV-106 fail open • TIC-106 LO 30 / HI 48 °C • TI-109 HI 130 °C • TSHH-109 at 150 °C for 3 s → I-1	S4 L3 **12 High**

HAZOP recommendation
Installed spare cooling-medium pump with auto-start on P-160 failure.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R28

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R28**
Node	N4 / N6 / N10 · Intercooler E-101 / Aftercooler E-102 / Cooling medium	**Action party**	Instrument
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.1	Temperature **MORE**: Section 2 suction hot	• E-101 fouling (fault library) • TV-106 throttled by a false low reading (fault library: TT-106 reads low) or TIC-106 in MAN low • cooling medium hot or lost (node N10)	Hot gas into section 2: its discharge temperature, power and seal temperatures rise. With TV-106 shut and no trip the model's TI-109 passes 150 °C after 95 s and peaks at 193 °C.	• TIC-106 (SP 40 °C) → TV-106 fail open • TIC-106 LO 30 / HI 48 °C • TI-109 HI 130 °C • TSHH-109 at 150 °C for 3 s → I-1	S3 L3 **9 Medium**
N6.3	Temperature **MORE**: Export gas hot	• TV-111 shut or TIC-111 in MAN low • cooling medium lost (N10) • E-102 fouling	Export gas above the pipeline coating limit (assumed 60 °C): coating damage, upheaval buckling of the line. With no cooling and no trip the model's export temperature reaches 202 °C.	• TIC-111 (SP 45 °C) → TV-111 fail open • TIC-111 LO 30 / HI 55 °C • TSHH-109 at 150 °C for 3 s → I-1	S3 L2 **6 Medium**
N10.3	Flow **LESS**: Cooling medium to one cooler throttled	• TIC-106 acting on a false low reading (fault library: TT-106 reads low) • TIC-106 or TIC-111 in MAN with a low output	The gas leaves that cooler hotter than shown; section 2 or the export runs hot (see N4, N6).	• TI-109 HI 130 °C • TSHH-109 at 150 °C for 3 s → I-1 • TIC-111 (SP 45 °C) → TV-111 fail open • TIC-111 LO 30 / HI 55 °C	S3 L3 **9 Medium**

HAZOP recommendation
Second temperature element on the cooler outlets (deviation alarm) so a single transmitter cannot move the cooling medium unseen.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R29

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R29**
Node	N3 / N4 / N5 / N10 · Section 1 (LP) and UV-103 recycle / Intercooler E-101 / Section 2 (HP) and UV-108 recycle / Cooling medium	**Action party**	Operations
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.6	Temperature **MORE**: Section 1 discharge hot	• Section 1 fouling (fault library) • hot suction • high ratio • recycle at high flow	Discharge above the 150 °C sealing ceiling: seal and O-ring damage, gas leak at the shaft seals.	• TI-104 HI 125 °C • TSHH-104 at 150 °C for 3 s → I-1	S3 L3 **9 Medium**
N4.1	Temperature **MORE**: Section 2 suction hot	• E-101 fouling (fault library) • TV-106 throttled by a false low reading (fault library: TT-106 reads low) or TIC-106 in MAN low • cooling medium hot or lost (node N10)	Hot gas into section 2: its discharge temperature, power and seal temperatures rise. With TV-106 shut and no trip the model's TI-109 passes 150 °C after 95 s and peaks at 193 °C.	• TIC-106 (SP 40 °C) → TV-106 fail open • TIC-106 LO 30 / HI 48 °C • TI-109 HI 130 °C • TSHH-109 at 150 °C for 3 s → I-1	S3 L3 **9 Medium**
N5.3	Temperature **MORE**: Section 2 discharge hot	• Hot section 2 suction (node N4) • section 2 fouling (fault library) • high ratio from a high arrival pressure	Discharge above the 150 °C sealing ceiling: seal damage and gas leak.	• TI-109 HI 130 °C • TSHH-109 at 150 °C for 3 s → I-1	S3 L3 **9 Medium**
N5.5	Pressure **MORE**: Section ratio split shifts	Section 1 or section 2 fouling (fault library)	One section takes more ratio: hotter discharge, more power, smaller margin in that section.	• TI-104 HI 125 °C • TI-109 HI 130 °C • SIC-100 LO 72 / HI 104 %	S2 L3 **6 Medium**
N10.2	Temperature **MORE**: Cooling medium supply hot	• Seawater exchanger fouled (fault library: cooling medium supply hot) • hot seawater	TV-106/TV-111 open fully and the coolers lose control: hotter section 2 and export, more power.	• TI-160 cooling-medium supply indication (no alarm) • TIC-106 (SP 40 °C) → TV-106 fail open • TIC-106 LO 30 / HI 48 °C • TIC-111 (SP 45 °C) → TV-111 fail open • TIC-111 LO 30 / HI 55 °C • TI-109 HI 130 °C	S2 L3 **6 Medium**

HAZOP recommendation

Cooler performance monitoring (E-101/E-102 duty and approach, TV positions) and a cleaning programme for the coolers and the seawater exchanger.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R30

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R30**
Node	N10 · Cooling medium	**Action party**	Process
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N10.5	Pressure **MORE**: Gas into the cooling-medium circuit	Tube rupture in E-101 (60 barg) or E-102 (121 barg)	High-pressure gas enters the low-pressure cooling-medium circuit: exchanger shell and P-160 overpressure, gas released at the expansion tank.	Cooling-medium system design (separate sheet; assumed)	S4 L2 **8 Medium**
N10.6	Composition **AS WELL AS**: Gas in the cooling-medium return	Small tube leak	Gas at the expansion tank vent, P-160 vapour-locked (then cooling lost).	Cooling-medium system design (separate sheet; assumed)	S3 L2 **6 Medium**

HAZOP recommendation

Tube-rupture case for E-101 and E-102 (API 521): relief on the cooling-medium side and gas detection at the cooling-medium expansion tank.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R31

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R31**
Node	N11 · Blowdown BDV-101 / BDV-112	**Action party**	Instrument / Safety
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N11.1	Flow **NO**: Blowdown fails	BDV stuck shut, solenoid fails energised, restriction orifice blocked by hydrate	The loop stays at the settle-out pressure (≈ 69 barg) during a fire: vessel or pipe ruptures when the steel weakens; escalation.	• ESD-2: BDV-101 and BDV-112 open to flare (fail open, RO 45 mm, choked), XV-001 shuts the wells • BDV OPEN alarm • PSV-107 80 barg on V-102 (KC-PID-100 note 3: on hold, not drawn — not credited until installed)	S5 L2 **10 High**

HAZOP recommendation
BDV-101/BDV-112 proof test (solenoid, stroke, restriction orifice) and confirmation of the 15-min blowdown with the fire case.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R32

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R32**
Node	N2 / N11 · Inlet scrubber V-101 / Blowdown BDV-101 / BDV-112	**Action party**	Materials
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.8	Temperature **LESS**: Cold gas when pressurising	Pressurising a depressurised loop quickly through HV-101	Joule–Thomson cooling across HV-101 of about 13 K (0.45 K/bar from about 2 to 30 barg): hydrates in HV-101 and V-101, low-temperature embrittlement of the valve body.	Written procedures (Exercises tab: cold start-up, normal shutdown, ESD with blowdown, GT trip restart)	S2 L3 **6 Medium**
N11.3	Temperature **LESS**: Cold flare line during blowdown	Expansion cooling of the gas through the BDVs and in the emptying vessels	Flare line and BDV outlets cool to about −21 °C in the model: brittle fracture of carbon steel, hydrates in the orifices.	Flare lines LTCS, MDMT −46 °C (KC-PID-100 note 3, TBC)	S4 L2 **8 Medium**

HAZOP recommendation
Low-temperature materials: confirm flare lines, BDV/HV-102 outlets and HV-101 for the expansion cooling (MDMT −46 °C, note 3, TBC).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R33

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R33**
Node	N11 · Blowdown BDV-101 / BDV-112	**Action party**	Process
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N11.7	Composition **AS WELL AS**: Liquid to flare	Scrubber liquids carried through the BDVs or PV-102	Burning liquid rain from the flare tip on the platform or the sea.	Flare knock-out drum (outside the scope; assumed)	S4 L2 **8 Medium**
N11.8	Ignition **NO**: Flare not lit	Pilot flame-out, ignition system failure	Unburnt gas cloud from the flare tip drifting over the platform: explosion if it finds an ignition source.	Flare pilots (assumed)	S4 L2 **8 Medium**

HAZOP recommendation
Flare knock-out drum, flare pilot monitoring and liquid carry-over protection (outside this node set; confirm in the flare study).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R34

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R34**
Node	N11 / N12 · Blowdown BDV-101 / BDV-112 / Procedures	**Action party**	Operations / Safety
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N11.5	Timing **LATE**: Blowdown started late	• ESD-2 is a manual decision • operator unsure or busy with the trip	A fire on the module heats a pressurised loop: escalation before the inventory is removed.	• Operator ESD-1 / ESD-2 on the SIS screen • Fire and gas detection and ESD (not on KC-PID-100; assumed)	S4 L2 **8 Medium**
N12.9	Blowdown **OTHER THAN**: ESD-2 used where ESD-1 was meant	• Wrong push-button • unclear criteria for blowing down	Unnecessary blowdown: the whole loop inventory flared, cold flare lines, a longer restart (re-pressurising from about 0 barg).	• Operator ESD-1 / ESD-2 on the SIS screen • Written procedures (Exercises tab: cold start-up, normal shutdown, ESD with blowdown, GT trip restart)	S2 L3 **6 Medium**

HAZOP recommendation
ESD-2 criteria in the emergency procedure (when to blow down) and automatic ESD-2 from confirmed fire detection.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R35

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R35**
Node	N7 / N8 / N12 · Export riser and pipeline to the onshore terminal / GT-100 driver and fuel gas / Procedures	**Action party**	Operations / Process
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N7.2	Flow **NO**: Onshore intake shut	• Terminal inlet ESD valve closes • terminal blackout	The pipeline packs at the full export rate; PSHH-112 trips K-100 and production is flared until the wells are cut back.	• PI-112 HI 126 barg • PSHH-112 at 132 barg for 2 s → I-1 • Trip feed-forward: on I-1 PIC-102 output goes to ≥ 80 % (production flared at once)	S3 L2 **6 Medium**
N8.1	Power **NO**: Loss of the driver	• GT-100 flame-out (fault library) • fuel valve closes (fail closed on air loss) • GT internal protection	K-100 coasts down (at rest after about 7 min in the model); surge during the rundown unless both recycles open; the whole production (222 t/h) goes to flare until the wells are cut back; export lost.	• GT-100 flame-out trips I-1 • XV-100 shuts the fuel • On I-1 UV-103 and UV-108 go to 100 % (trip opening) • Trip feed-forward: on I-1 PIC-102 output goes to ≥ 80 % (production flared at once) • PSHH-001 at 34.5 barg for 3 s → XV-001 shuts the wells	S3 L4 **12 High**
N12.7	Shutdown **AFTER**: Compressor stopped before the wells are cut back	ESD-1 used before production is reduced	The full production (222 t/h) goes to flare (lost sales, CO_2, radiation).	• Written procedures (Exercises tab: cold start-up, normal shutdown, ESD with blowdown, GT trip restart) • Trip feed-forward: on I-1 PIC-102 output goes to ≥ 80 % (production flared at once)	S2 L3 **6 Medium**
N12.8	Timing **LATE**: Slow recovery after a trip	Venting, reset and restart take hours	Production flared at about 222 t/h for the whole outage unless the wells are cut back.	Written procedures (Exercises tab: cold start-up, normal shutdown, ESD with blowdown, GT trip restart)	S2 L4 **8 Medium**

HAZOP recommendation
Automatic production cut-back signal to the wells on a K-100 trip, and ramp-rate limits for production changes agreed with production control.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R36

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R36**
Node	N2 / N12 · Inlet scrubber V-101 / Procedures	**Action party**	Operations
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.5	Flow **REVERSE**: Settled-out gas back into the header	ESDV-101 or HV-101 opened after a trip before the loop is vented	Gas at the settle-out pressure (≈ 69 barg) flows back into the 30-barg header: header overpressure, PSV-001 lifts.	• ESDV-101 open permissive: ΔP across the valve < 3 bar (pressurise with HV-101 first) • PSHH-001 at 34.5 barg for 3 s → XV-001 shuts the wells • PSV-001 set 36 barg on the production header → flare • Written procedures (Exercises tab: cold start-up, normal shutdown, ESD with blowdown, GT trip restart)	S3 L2 **6 Medium**
N12.3	Timing **EARLY**: Loaded before the anti-surge loops are in AUTO	SIC-100 put in cascade or speed raised with UIC-103/108 still in MAN	Surge on loading: thrust reversal, seal damage.	• SURGE alarms UIC-103 / UIC-108 (flow reversal) • VSHH-105 at 90 µm for 3 s → I-1 • VSHH-110 at 90 µm for 3 s → I-1	S3 L3 **9 Medium**
N12.5	Restart **OTHER THAN**: Restart without venting the settled-out loop	HV-101 or ESDV-101 opened at settle-out pressure instead of venting with HV-102 first	Gas at ≈ 69 barg back into the 30-barg header: PSV-001 lifts, PSHH-001 shuts the wells.	• ESDV-101 open permissive: ΔP across the valve < 3 bar (pressurise with HV-101 first) • PSHH-001 at 34.5 barg for 3 s → XV-001 shuts the wells • PSV-001 set 36 barg on the production header → flare	S3 L3 **9 Medium**

HAZOP recommendation
Restart procedure: vent the settled-out loop with HV-102 before HV-101 or ESDV-101 is opened; simulator training on restart and loading.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R37

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R37**
Node	N1 / N6 / N8 / N9 · Production header / Aftercooler E-102 / GT-100 driver and fuel gas / Lube oil and seal system	**Action party**	Safety
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.9	Integrity **OTHER THAN**: Leak from the header	Flange or small-bore failure, PSV-001 inlet, corrosion	Gas release at about 30 barg on the compressor module: jet fire or explosion if ignited.	• Fire and gas detection and ESD (not on KC-PID-100; assumed) • Operator ESD-1 / ESD-2 on the SIS screen	S4 L2 **8 Medium**
N6.7	Integrity **OTHER THAN**: Leak at 120 barg	• Flange, FE-114 or small-bore failure • vibration fatigue	High-pressure gas jet on the module: jet fire, explosion, escalation to the riser.	• Fire and gas detection and ESD (not on KC-PID-100; assumed) • Operator ESD-1 / ESD-2 on the SIS screen • ESD-2: BDV-101 and BDV-112 open to flare (fail open, RO 45 mm, choked), XV-001 shuts the wells	S5 L2 **10 High**
N8.7	Integrity **OTHER THAN**: Fuel-gas leak in the GT enclosure	Fuel-line flange or fitting failure, vibration	Gas accumulates in the enclosure: explosion.	Fire and gas detection and ESD (not on KC-PID-100; assumed)	S4 L2 **8 Medium**
N9.6	Flow **MORE**: Seal leakage high	Shaft-seal damage by surge thrust, liquid carry-over or discharge above 150 °C	Process gas to the seal vent or into the compressor house: fire or explosion.	• Seal-gas panel with its own alarms (separate sheet, not modelled; assumed) • TSHH-104 at 150 °C for 3 s → I-1 • TSHH-109 at 150 °C for 3 s → I-1	S4 L2 **8 Medium**

HAZOP recommendation
Cover gas release, fire and gas detection and the ESD response of the compressor module in the fire-and-gas study (not on KC-PID-100).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R38

Project	Gas Export Compressor Ver 1 · K-100 with GT-100 driver · Operator Training Simulator	**Document**	KC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	**Action No.**	**R38**
Node	N1 / N3 · Production header / Section 1 (LP) and UV-103 recycle	**Action party**	Instrument / Process
Drawings	KC-PID-100 rev A (P&ID tab); KC-PFD-100 rev A (PFD tab)	**Due date**	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.1	Flow **NO**: No production gas	• Wells shut in by production control or by XV-001 (PSHH-001, ESD-2) • HP separator trip upstream	Suction pressure falls while K-100 runs: the machine slows to minimum speed and recycles; below the PSLL-101 setting it trips; export stops and the pipeline line-pack falls (sales lost).	• PIC-101 LO 28.5 / HI 31.5 barg • UIC-103 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-103 cooled recycle, fail open • PSLL-101 at 20 barg for 5 s → I-1 (armed in RUN)	S2 L3 **6 Medium**
N1.2	Flow **LESS**: Production falls fast	• Wells cut back (fault library: production falls by up to 55 %) • well work • choke change upstream	K-100 turns down; both operating points move toward surge. If the anti-surge loops do not keep up the sections surge: flow reversal, thrust reversal, seal and bearing damage, possible gas release at the seals.	• UIC-103 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-103 cooled recycle, fail open • UIC-108 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-108 cooled recycle, fail open • UIC-103 margin LO 7 % and SURGE alarm • SURGE alarms UIC-103 / UIC-108 (flow reversal) • VSHH-105 at 90 µm for 3 s → I-1	S4 L2 **8 Medium**
N3.1	Flow **LESS**: Section 1 driven into surge	• Fast production cut • UV-103 sticks (fault library) • anti-surge loop too slow or badly tuned	Surge: forward flow collapses and reverses every few seconds, thrust reverses, temperature builds up in the loop; seals and thrust bearing fail, rotor rub, gas release at the seals.	• UIC-103 anti-surge (control line 10 % margin, fast opening below 5 %) → UV-103 cooled recycle, fail open • UIC-103 margin LO 7 % and SURGE alarm • VSHH-105 at 90 µm for 3 s → I-1 • TSHH-104 at 150 °C for 3 s → I-1	S4 L3 **12 High**

HAZOP recommendation
Feed-forward of the production flow FI-001 to UIC-103/UIC-108 (or rate-of-change limits) so a fast production cut does not drive the margins below the fast-opening line.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☑ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Asset / project manager			